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Imam et al.

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(54) **METHODS FOR PRODUCING A COMPOSITION CONTAINING LIPIDS FROM RECOMBINANT ALGAE HAVING HIGH LIPID PRODUCTIVITY**

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Related U.S. Application Data

(63) Continuation of application No. 17/519,383, filed on Nov. 4, 2021, now Pat. No. 11,976,315.

(60) Provisional application No. 63/110,301, filed on Nov. 5, 2020.

(51) **Int. Cl.**
C12P 7/64 (2022.01)
C07K 14/405 (2006.01)
C12N 1/12 (2006.01)

(52) **U.S. Cl.**
CPC **C12P 7/64** (2013.01); **C07K 14/405** (2013.01); **C12N 1/125** (2021.05)

(58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

The invention provides recombinant algal mutants that have a genetic modification to a gene or nucleic acid sequence encoding a WD40 repeat containing protein or domain. The genetic modification of one or more nucleic acid sequences encoding a WD40 repeat containing protein or domain results in a mutant organism with increased lipid productivity and/or higher biomass productivity (as measured by total organic carbon). The genetic modification can be a gene attenuation or functional deletion. The lipid products of these mutants can be utilized as biofuels or for other specialty chemical products. Methods of making and using the recombinant algal mutants and methods of producing lipids are also disclosed.

18 Claims, 8 Drawing Sheets

Specification includes a Sequence Listing.

FIG. 1A

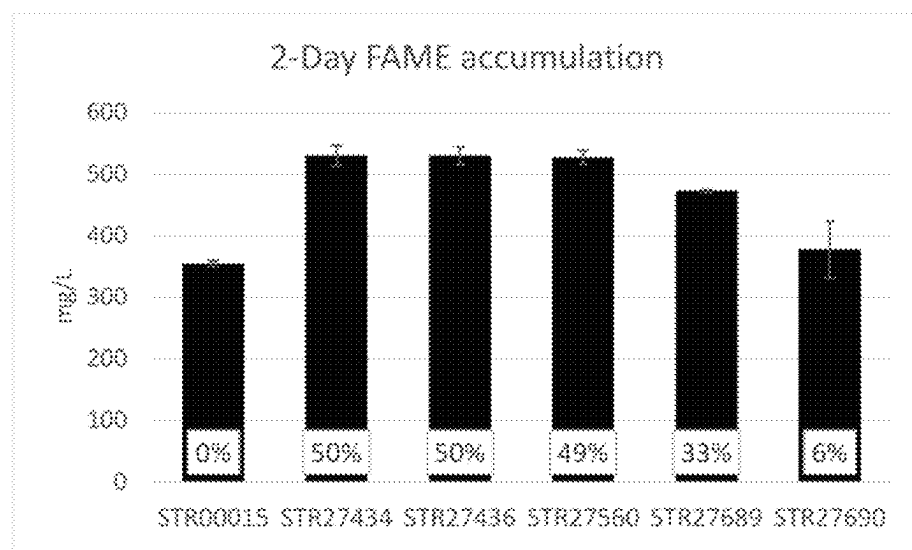


FIG. 1B

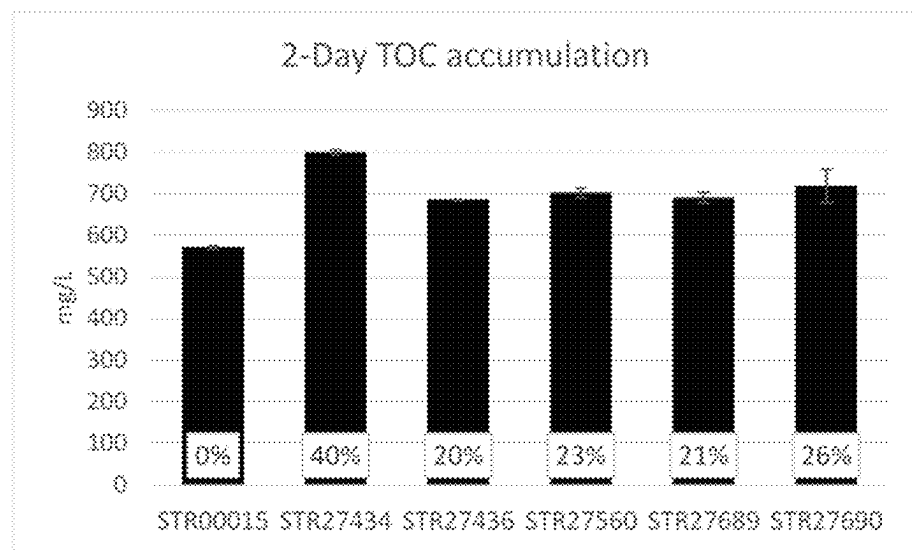


FIG. 1C

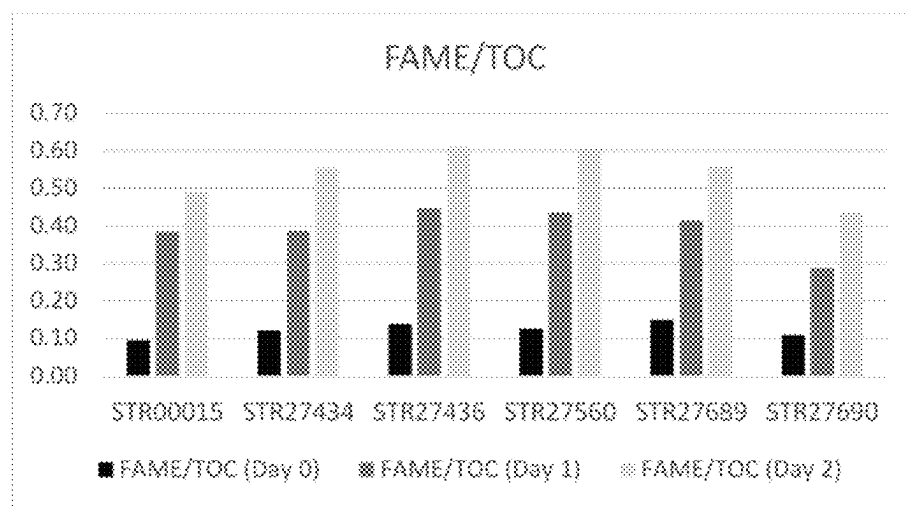


FIG. 2A

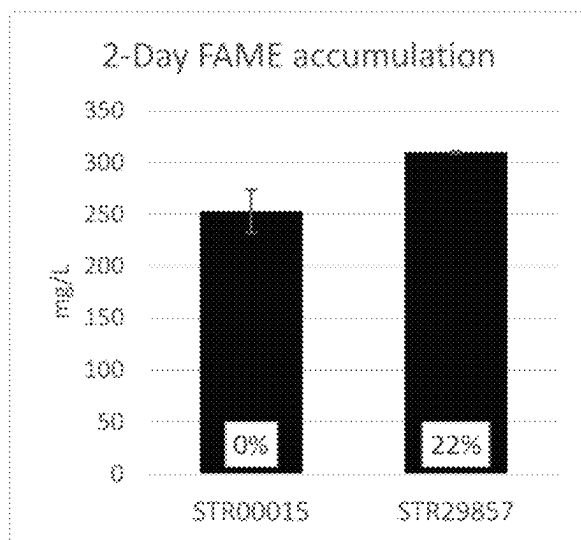


FIG. 2B

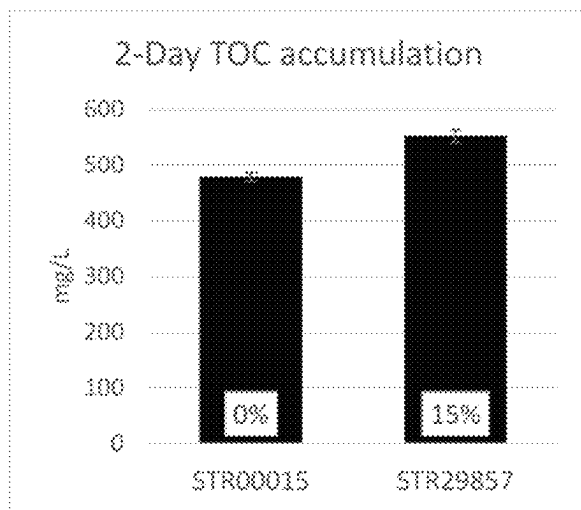


FIG. 2C

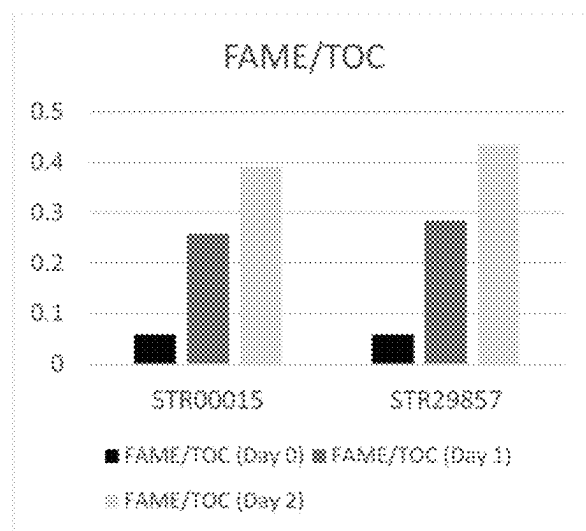


FIG. 3A

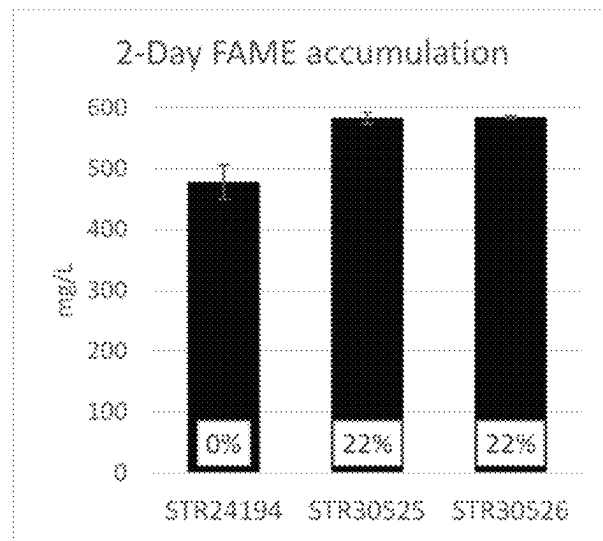


FIG. 3B

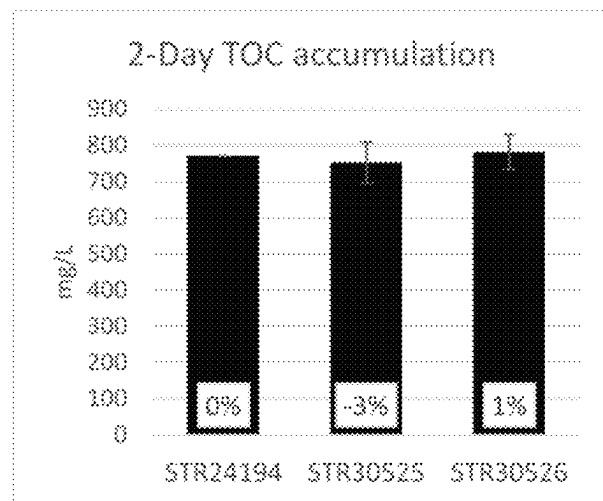


FIG. 3C

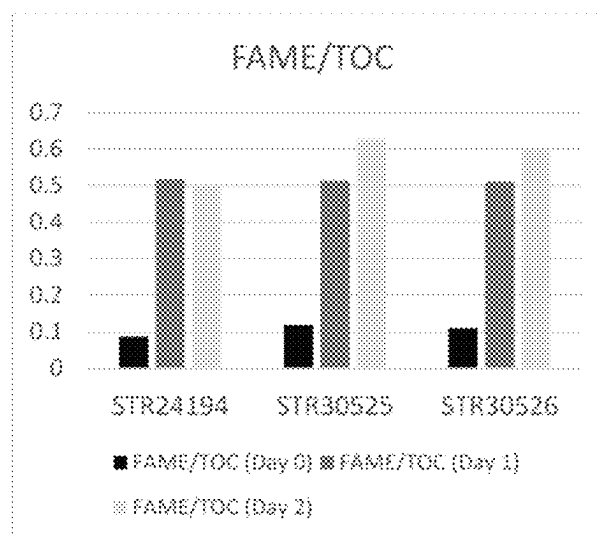


FIG. 4A

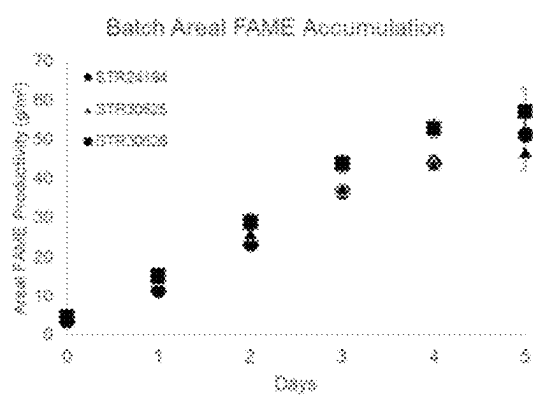


FIG. 4B

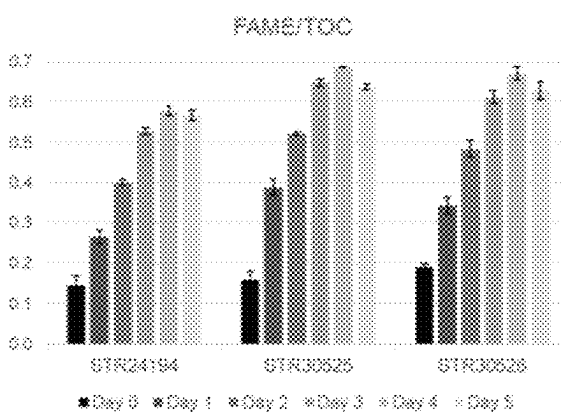
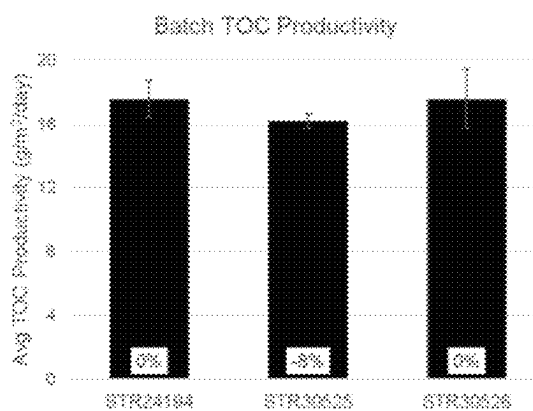
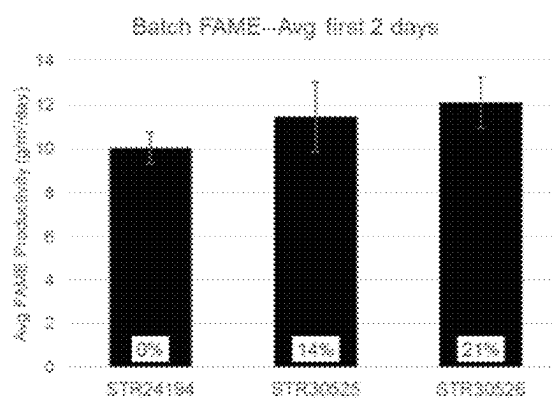


FIG. 4C

FIG. 4D

[illegible]

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**METHODS FOR PRODUCING A
COMPOSITION CONTAINING LIPIDS FROM
RECOMBINANT ALGAE HAVING HIGH
LIPID PRODUCTIVITY**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation application of U.S. application Ser. No. 17/519,383 filed Nov. 4, 2021, now pending; which claims the benefit under 35 USC § 119(e) to U.S. Application Ser. No. 63/110,301 filed Nov. 5, 2020. The disclosure of each of the prior applications is considered part of and is incorporated by reference in the disclosure of this application.

**INCORPORATION BY REFERENCE OF
SEQUENCE LISTING**

The material in the accompanying sequence listing is hereby incorporated by reference into this application. The accompanying sequence listing xml file, named SGI2290-2_ST26.xml, was created on Mar. 29, 2024 and is 73,099 bytes in size.

BACKGROUND OF THE INVENTION

Field of the Invention

The invention involves the provision of a recombinant algal mutant organism and methods for the production of lipids.

Background Information

The production of biofuels presents great opportunities to develop environmentally sound sources of energy that can be obtained at reasonable cost. Efforts have been directed towards using algae or other microorganisms to produce hydrocarbons that can be used as biodiesel or other biofuels due to their high lipid content. Additional specialty chemicals can also be obtained from these organisms and for use in consumer products.

Since algae use energy from sunlight to combine water and carbon dioxide to produce biomass, achieving increased productivity offers the possibility of a carbon neutral fuel source. The development of algal strains with very high lipid productivity for the production of algal-sourced biofuels therefore presents the possibility of a significant reduction in new carbon dioxide released into the atmosphere and a consequent reduction in the problem of global warming.

The development of commercially viable algal biofuels requires strains with high lipid and biomass productivity. Even the most productive wild type strains are not sufficiently productive to permit an economically viable development of this resource. Strategies for increasing algal production of biofuels and other products have included modification of nutrition provided to the organisms, such as cultivating the organisms in nitrogen, phosphorus, or silicon deficient media. Other strategies have included modification of cultivation conditions or environmental protocols, or various efforts directed towards genetic engineering of the organisms. While engineering algae strains to have a combination of increased photosynthetic efficiency (resulting in increased overall biomass productivity) and/or high lipid productivity could provide a solution to this problem, defi-

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ciencies still remain. The development of more productive strains continues to be a barrier to efficient utilization of this energy source.

SUMMARY OF THE INVENTION

The invention provides recombinant algal mutants that have a genetic modification to a gene or nucleic acid sequence encoding a WD40 repeat containing protein. The genetic modification of one or more nucleic acid sequences encoding a WD40 repeat containing protein results in a mutant organism with increased lipid productivity and/or higher biomass productivity (as measured by total organic carbon). In some embodiments the genetic modification is a gene attenuation, or a deletion, disruption, or inactivation. The lipid products of these mutants can be utilized as biofuels or for other specialty chemical products. Methods of making and using the recombinant algal mutants and methods of producing lipids are also disclosed.

In a first aspect the invention provides a recombinant algal organism having a genetic modification in a gene or nucleic acid sequence encoding a WD40 repeat containing protein. The recombinant alga exhibits higher lipid productivity and/or higher biomass productivity versus a corresponding control algal cell not having the genetic modification. In one embodiment the recombinant alga is a Chlorophyte alga, which optionally can be of the Class Trebouxiophyceae. The gene or nucleic acid sequence encoding the WD40 repeat containing protein can have a nucleic acid sequence having at least 75% sequence identity to a nucleic acid sequence of any one of SEQ ID NOs: 14-26. In one embodiment the gene encoding the WD40 repeat containing protein has a nucleic acid sequence having at least 75% sequence identity to a nucleic acid sequence of SEQ ID NO: 15 or 16.

In one embodiment the genetic modification to the gene encoding the WD40 repeat containing protein is a functional deletion. The genetic modification of the gene encoding the WD40 repeat containing protein can result in an attenuation in the expression of the nucleic acid sequence. In one embodiment the genetic modification of the gene encoding the WD40 repeat containing protein is a deletion of one or more amino acids in a polypeptide sequence of a WD40 repeat containing protein. In one embodiment the genetic modification of the gene encoding the WD repeat containing protein is a frame shift mutation. The genetic modification can also be a deletion, a disruption, or an inactivation.

In one embodiment the recombinant alga has at least 30% higher lipid productivity versus a corresponding control alga that does not comprise the genetic modification. The recombinant alga can have at least 15% higher biomass productivity as measured by total organic carbon versus a control alga that does not comprise the genetic modification. The recombinant alga can exhibit at least 40 grams per square meter per day of lipid production after 5 days of cultivation.

In various embodiments the recombinant alga can higher biomass productivity per unit time as measured by production of total organic carbon (TOC). The recombinant alga can have higher biomass productivity under nitrogen deficient conditions. The recombinant alga can have higher total organic carbon production under nitrogen deficient conditions. In some embodiments the recombinant alga is of a family selected from Oocystaceae, Chlorrellaceae, and Eustigmatophyceae; and the recombinant alga can be of a genus selected from *Chlorella*, *Parachlorella*, *Picochlorum*, *Tetraselmis*, and *Oocystis*.

In another aspect the invention provides a lipid containing product produced by any of the recombinant algae described herein.

In another aspect the invention provides a biomass product containing any of the recombinant algae described herein.

In another aspect the invention provides a method of producing a composition containing lipids. The methods involve cultivating a recombinant alga described herein and thereby producing a composition containing lipids. The method can also involve harvesting a lipidic composition from the recombinant alga. The method can involve producing a genetic modifications to a nucleic acid sequence encoding a WD repeat protein, which can be a functional deletion. The functional deletion can be obtained by subjecting the algal organism to ultra-violet light. In some embodiments the gene encoding the WD repeat containing protein has a nucleic acid sequence having at least 90% sequence identity to SEQ ID NO: 16. The algal organism can be a Chlorophyte alga, optionally of the genus *Oocystis*, but can be any recombinant alga described herein. The recombinant alga produced by the method can have at least 30% greater lipid productivity versus a control alga.

In another aspect the invention provides a method of producing a composition containing lipids. The methods involve performing a genetic modification to an algal organism in a gene encoding a WD40 repeat containing protein, and culturing the organism to thereby produce a composition containing lipids. The method can further involve harvesting a lipidic composition from the algal organism.

In another aspect the invention provides a method for identifying a recombinant algal organism with high lipid productivity and/or high biomass productivity. The method involves mutagenizing a population of algal organisms; screening the mutagenized algal organisms for higher lipid productivity; sequencing at least a portion of the genome of the mutagenized algal organisms; identifying genetic changes in the mutagenized organisms compared to the population of algal organisms prior to mutagenesis; recapitulating the genetic changes in a parental strain of the mutagenized algal organisms; and thereby identifying a recombinant algal organism having high lipid productivity. The method can further involve harvesting a lipidic composition from the recombinant algal organism. The recombinant algal organism with high lipid productivity and/or high biomass productivity can be any recombinant algal organism described herein. In one embodiment the recombinant algal organism can have at least 40% greater lipid productivity versus a control alga.

The summary of the invention described above is not limiting and other features and advantages of the invention will be apparent from the following detailed description of the invention, and from the claims. Section headings or sub-headings are provided solely for the convenience of the reader, and do not denote a departure from discussion or necessarily an entirely new subject matter area. Any subject matter can be discussed or disclosed under any section heading or sub-heading.

DETAILED DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1C are graphical illustrations of the effects on FAME and TOC accumulation in strains having a genetic modification to a nucleic acid sequence encoding a WD40 repeat family protein by mutagenesis of wild type cells (Strain 15). FIG. 1A shows 2-day FAME accumulation for various *Oocystis* strains. FIG. 1B shows 2-day TOC accu-

mulation. FIG. 1C shows the ratio of FAME/TOC for various strains as an indicator of carbon partitioning. Mutagenized strains include 27434, 27436, 27550, 27689, and 27690, which all contain a deletion to SEQ ID NO: 1 (a WD40 repeat containing protein). The 2-day data was measured after 48 hours of growth in nitrogen deplete media.

FIGS. 2A-2C are graphical illustrations of the FAME and TOC accumulation in wild type strains constructed to have a deletion of a nucleic acid sequence encoding a WD40 repeat family protein. FIG. 2A shows 2-day FAME accumulation for modified *Oocystis* sp. strains (29857) versus the wild type (Strain 15). FIG. 2B shows 2-day TOC accumulation. FIG. 2C shows the ratio of FAME/TOC, a useful measure of carbon partitioning.

FIGS. 3A-3C are graphical illustrations of the FAME and TOC accumulation in a laboratory strain (24194) constructed to have a deletion of a nucleic acid sequence encoding a WD40 repeat family protein versus the (control) laboratory strain. FIG. 3A shows 2-day FAME accumulation for the laboratory strain (Strain 24194) and modified *Oocystis* sp. strains (30525 and 30526). FIG. 3B shows 2-day TOC accumulation. FIG. 3C shows the ratio of FAME/TOC, a useful measure of carbon partitioning.

FIGS. 4A-4D provide a graphical illustration of 5-day data for FAME and TOC productivity under nitrogen deplete conditions. FIG. 4A shows areal FAME productivity v. days. FIG. 4B shows average FAME productivity for the first 2 days. FIG. 4C shows TOC productivity. And FIG. 4D shows the FAME/TOC ratios over a 5 day period.

FIG. 5 is a sequence alignment of protein sequences of various Chlorophyte algae species. It is shown that there is a high degree of sequence identity across Chlorophyte species in key domains of the polypeptide sequence.

DETAILED DESCRIPTION OF THE INVENTION

The invention provides recombinant algal mutants that have a genetic modification to a nucleic acid sequence encoding a WD40 repeat containing protein. The genetic modification(s) described herein result in a recombinant or mutant cell or organism with higher lipid productivity and/or higher biomass productivity. The recombinant algal mutants can also optionally have reduced chlorophyll content and/or a reduced PSII antenna size versus a corresponding control cell or organism not having the genetic modification. In various embodiments the genetic modification(s) described herein can result in substantial increases in lipid productivity and/or biomass productivity. Any of the recombinant cells or organisms disclosed herein can be mutant photosynthetic organisms.

The recombinant cell or organism of the invention having a genetic modification described herein can have higher lipid productivity and/or higher biomass productivity than a corresponding (control) cell or organism. In some embodiments the genetic modification is an attenuation of a gene or nucleic acid sequence encoding a WD40 repeat containing protein. In any embodiment lipid productivity can be measured by FAME accumulation, and biomass productivity can be measured as the rate of biomass accumulation, for example as the total organic carbon (TOC) content of the respective cells or organisms.

In one embodiment the mutant or recombinant algal cell or organism has higher lipid and/or biomass productivity in batch culture compared to a corresponding (control) cell or organism not having the genetic modification. Batch culture is a culture where nutrients are not renewed or re-supplied

to the medium during the time period the cells or organisms are cultured. In other embodiments the mutant or recombinant algal cell or organism has higher lipid and/or biomass productivity under nitrogen deplete conditions compared to a corresponding (control) cell or organism. In some embodiments the higher lipid and/or biomass productivity is achieved under batch culture and nitrogen deplete conditions; and in other embodiments the higher lipid and/or biomass productivity is achieved under semi-continuous or continuous culture conditions. Semi-continuous culture conditions are conditions where a fixed volume of culture is removed at regular time intervals and an equal volume of fresh medium is immediately added back to the culture. In various embodiments the fixed volume removed and added is from 10% to 90% of the total culture volume. Continuous culture conditions involve regular removal of culture with immediate replacement with fresh medium. Persons of ordinary skill realize that time intervals can vary with the purpose of the culture and continuous culture conditions involve a more frequent removal of culture and replacement with fresh medium than semi-continuous conditions.

Any of the mutant or recombinant cells or organisms disclosed herein can be photosynthetic cells or organisms. Any of the mutant or recombinant cells or organisms described herein can exhibit increased lipid productivity and/or increased biomass productivity under photoautotrophic conditions compared to a corresponding control cell or organism, i.e., conditions where the recombinant cells or organisms can produce their own biomass using light, carbon dioxide, water, and nutrients via photosynthesis. Corresponding (control) cells or organisms are useful for evaluating the effect of any one or more of the genetic modifications. Corresponding (control) cells or organisms do not have the one or more genetic modifications being evaluated and are subjected to the same or substantially the same conditions as the test cells or organisms such that a difference in the performance or characteristics of the cells or organisms is based only on the genetic modification(s) being evaluated. In any embodiment the corresponding (control) cells or organisms can be of the same species as the test organism. They can be derived from the same parent cell or parental line. They can also be the same or similar in every way except for the one or more genetic modification(s) being evaluated. In some embodiments the corresponding (control) cell or organism is a wild-type cell or organism. But the corresponding (control) cell or organism can also be a laboratory strain or parental strain of the test cell or organism. Substantially the same conditions can be the same conditions or slightly different conditions where the difference does not materially affect the function, activity, or expression of the nucleic acid sequence modified.

In various embodiments the recombinant cells or organisms are algal cells. In one embodiment the recombinant alga has a genetic modification to a gene encoding a WD40 repeat containing protein. The lipid products of these mutants can be further processed into biofuels or used in the production of other specialty chemical products. The gene or nucleic acid sequence encoding the WD40 repeat containing protein or domain that contains the genetic modification can encode any of the polypeptide sequences described herein, hereby disclosed in all possible combinations or sub-combinations as if set forth fully herein.

In some embodiments recombinant cells or organisms of the invention can have a reduced amount of chlorophyll b, and can have an increased chlorophyll a to chlorophyll b ratio (chl a/chl b) compared to a corresponding control cell or organism. The recombinant cells or organisms can have

decreased photosynthetic antenna size, for example reduced photosystem II (PSII) and/or reduced photosystem I (PSI) antenna size. In various embodiments the cross-sectional unit size of the PSII and/or PSI antenna of the recombinant cells or organisms disclosed herein can be reduced by at least 10%, at least 20%, at least 30%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, or at least 60% compared to the PSII and/or PSI antenna size of a corresponding control cell or organism.

In any embodiment the recombinant cells or organisms can have a higher growth rate and/or a higher biomass productivity than a corresponding control cell or organism not having the genetic modification, for example, higher biomass productivity per hour or per day or per period of 2 days or 3 days or 4 days or 5 days or 6 days. "Biomass" refers to cellular mass, whether of living or dead cells. Biomass productivity, or biomass accumulation, or growth rate, can be measured by any means accepted in the art, for example as ash free dry weight (AFDW), dry weight, wet weight, or total organic carbon (TOC) productivity. In any embodiment biomass productivity, or biomass accumulation, or the growth rate, can be measured as total organic carbon (TOC) productivity.

The recombinant cells or organisms of the invention can produce a greater amount of a bioproduct per time period (e.g., per minute or per hour or per day or per period of 2 days or 3 days or 4 days or 5 days or 6 days), for example a lipid product (which can optionally be measured as FAME), a fatty acid product, a carbohydrate, a protein product, or a polymer than a corresponding (control) organism not having the genetic modification(s). The amount of bioproduct can be expressed as g/time period, mg/time period, ug/time period, or any other defined quantity per defined time period described herein. Such bioproducts can be isolated from a lysate or biomass or cellular secretion of any of the recombinant cells or organisms of the invention. In some embodiments, the recombinant cells or organisms of the invention produce at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 100% more of a bioproduct than a corresponding control alga cultured under the substantially the same conditions, which can be batch, semi-continuous, or continuous culture conditions and may be nutrient replete culture conditions or may be nitrogen deplete conditions, and may be photoautotrophic conditions. In some embodiments the recombinant cells or organisms of the invention produce at least 20% or at least 30% more lipid (e.g., as measured by FAME) over a 2-day period, or at least 30% or at least 40% more of a lipid product over a 5-day period.

Without wanting to be bound by any particular theory it is believed that the genetic modifications described herein result in an attenuation or partial or complete elimination of expression of a WD40 repeat containing protein. Such attenuation or elimination results in a significant increase in lipid productivity in the cell, which in one embodiment can be measured as the total FAME produced by the cell. A further result can be a significant increase in biomass productivity, which in one embodiment can be demonstrated by the organic carbon produced by the cell (as measured, for example, by total organic carbon).

As used herein, "exogenous" with respect to a nucleic acid or gene indicates that the nucleic acid or gene has been introduced (e.g., "transformed") into an organism, microorganism, or cell by human intervention. For example, such an exogenous nucleic acid can be introduced into a cell or organism via a recombinant nucleic acid construct. An

exogenous nucleic acid can be a sequence from one species introduced into another species, i.e., a “heterologous” nucleic acid. A heterologous nucleic acid can also be an exogenous synthetic sequence not found in the species into which it is introduced. An exogenous nucleic acid can also be a sequence that is homologous to an organism (i.e., the nucleic acid sequence occurs naturally in that species or encodes a polypeptide that occurs naturally in the host species) that has been isolated and subsequently reintroduced into cells of that organism. In some embodiments an exogenous nucleic acid that includes a homologous sequence can be distinguished from the naturally-occurring sequence by the presence of non-natural sequences linked to the exogenous nucleic acid, which can include but are not limited to non-native regulatory sequences attached to the homologous gene sequence in a recombinant nucleic acid construct. Alternatively or in addition, a stably transformed exogenous nucleic acid can be detected and/or distinguished from a native gene by its juxtaposition to sequences in the genome where it has integrated. Further, a nucleic acid is considered exogenous if it has been introduced into a progenitor of the cell, organism, or strain under consideration.

A “recombinant” or “engineered” nucleic acid molecule is a nucleic acid molecule that has been altered through human manipulation. As non-limiting examples, a recombinant nucleic acid molecule includes any nucleic acid molecule that: 1) has been partially or fully synthesized or modified in vitro, for example, using chemical or enzymatic techniques (e.g., by use of chemical nucleic acid synthesis, or by use of enzymes for the replication, polymerization, digestion (exonucleolytic or endonucleolytic), ligation, reverse transcription, transcription, base modification (including, e.g., methylation), integration or recombination (including homologous and site-specific recombination) of nucleic acid molecules); 2) includes conjoined nucleotide sequences that are not conjoined in Nature; 3) has been engineered using molecular biology techniques such that it lacks one or more nucleotides with respect to the naturally occurring nucleic acid molecule sequence; and/or 4) has been manipulated using molecular biology techniques such that it has one or more sequence changes or rearrangements with respect to the naturally occurring nucleic acid sequence, or has a sequence (e.g., by insertion) not found in the naturally occurring nucleic acid sequence. As non-limiting examples, a cDNA is a recombinant DNA molecule, as is any nucleic acid molecule that has been generated by in vitro polymerase reaction(s), or to which linkers have been attached, or that has been integrated into a vector, such as a cloning vector or expression vector.

When applied to organisms, the terms “transgenic” “transformed” or “recombinant” or “engineered” or “genetically engineered” refer to organisms that have been manipulated by introduction of an exogenous or recombinant nucleic acid sequence into the organism, or by genetic modification of native sequences (which are therefore then recombinant). Recombinant or genetically engineered organisms can also be organisms into which constructs for gene “knock down,” deletion, attenuation, inactivation, or disruption have been introduced to perform the indicated manipulation. Such constructs include, but are not limited to, RNAi, microRNA, shRNA, antisense, and ribozyme constructs. A recombinant organism can also include those having an introduced exogenous regulatory sequence operably linked to an endogenous gene of the transgenic microorganism, which can enable transcription in the organism. Also included are organisms whose genomes have been

altered by the activity of meganucleases or zinc finger nucleases. A heterologous or recombinant nucleic acid molecule can be integrated into a genetically engineered/recombinant organism’s genome or, in other instances, not integrated into a recombinant/genetically engineered organism’s genome, or can be present on a vector or other nucleic acid construct. As used herein, “recombinant microorganism” or “recombinant host cell” includes progeny or derivatives of the recombinant microorganisms of the disclosure.

Any of the recombinant algal cells or organisms described herein can be generated by human activity, for example, by classical mutagenesis and/or genetic engineering, but can also be produced by any feasible mutagenesis method, including but not limited to exposure to UV light, CRISPR/Cas9, cre/lox, gamma irradiation, or chemical mutagenesis. Screening methods can be used to identify mutants having desirable characteristics (e.g., reduced chlorophyll and increased lipid and/or biomass productivity). Methods for generating mutants of algal organisms using classical mutagenesis, genetic engineering, and phenotype or genotype screening are known in the art.

Algal Cell or Organism

The recombinant algal cell or organism of the invention can be a mutant microalga, or a mutant photosynthetic organism, or a mutant green alga. The recombinant alga can be any eukaryotic microalga such as, but not limited to, a Chlorophyte, an Ochrophyte, or a Charophyte alga. In some embodiments the mutant microalga can be a Chlorophyte alga of the taxonomic Class Chlorophyceae, or of the Class Chlorodendrophyceae, or the Class Prasinophyceae, or the Class Trebouxiophyceae, or the Class Eustigmatophyceae. In some embodiments, the mutant microalga can be a member of the Class Chlorophyceae, such as a species of any one or more of the genera *Asteromonas*, *Ankistrodesmus*, *Carteria*, *Chlamydomonas*, *Chlorococcum*, *Chlorogonium*, *Chrysothrix*, *Dunaliella*, *Haematococcus*, *Monoraphidium*, *Neochloris*, *Oedogonium*, *Pelagomonas*, *Pleurococcus*, *Pyrobotrys*, *Scenedesmus*, or *Volvox*. In other embodiments the mutant microalga of the invention can be a member of the Order Chlorodendrales, or Chlorellales. In other embodiments, the mutant microalga can be a member of the Class Chlorodendrophyceae, such as a species of any one or more of the genera *Prasinocladus*, *Scherffelia*, or *Tetraselmis*. In further alternative embodiments, the mutant alga can be a member of the Class Prasinophyceae, optionally a species of any one or more of the genera *Ostreococcus* or *Micromonas*. Further alternatively, the mutant microalga can be a member of the Class Trebouxiophyceae, and optionally of the Order Chlorellales, and optionally a genera selected from any one or more of *Botryococcus*, *Chlorella*, *Auxenochlorella*, *Heveochlorella*, *Marinichlorella*, *Oocystis*, *Parachlorella*, *Pseudochlorella*, *Tetrachlorella*, *Eremosphaera*, *Franceia*, *Micractinium*, *Nannochloris*, *Picochlorum*, *Prototheca*, *Stichococcus*, or *Viridiella*, or any of all possible combinations or sub-combination of the genera. In another embodiment the recombinant alga can be a Chlorophyte alga of the Class Trebouxiophyceae and the family Coccomyxaceae, and the genus *Coccomyxa* (e.g., *Coccomyxa subellipsoidea*). Or of the family Chlamydomonadaceae and the genus *Chlamydomonas* (e.g., *Chlamydomonas reinhardtii*); or of the family Volvocaceae and the genus *Volvox* (e.g., *Volvox carteri*, *Volvox aureus*, *Volvox globator*).

In another embodiment the recombinant alga is a Chlorophyte alga of the Class Trebouxiophyceae, or Eustigmatophyceae, and can be of the Order Chlorellales or Chlorodendrales, and can be of the Family Oocystaceae, or

Chlorellaceae, or Monodopsidaceae, and optionally from a genus selected from any one or more of *Oocystis*, *Parachlorella*, *Picochlorum*, *Nannochloropsis*, and *Tetraselmis*, in all possible combinations and sub-combinations. The recombinant alga can also be from the genus *Oocystis*, or the genus *Parachlorella*, or the genus *Picochlorum*, or the genus *Tetraselmis*, or from any of all possible combinations and sub-combinations of the genera. In one embodiment the recombinant algal cell or organism is of the Class Trebouxiophyceae, of the Order Chlorellales, and optionally of the family Oocystaceae, and optionally can be of the genus *Oocystis*.

Genetic Modification

In various embodiments the recombinant alga of the invention can have a genetic modification disclosed herein to a gene encoding a WD40 repeat containing protein. In one embodiment the recombinant alga of the invention has a genetic modification to a gene encoding a WD40 repeat containing protein or domain. In one embodiment the genetic modification is to a native or endogenous sequence of the cell or organism.

A “genetic modification” applied in the invention can be any one or more of an attenuation, a deletion, a gene “knock out,” a mutation, a disruption, an insertion, insertion of a stop codon, an inactivation, a rearrangement, one or more point mutations, a frameshift mutation, a nonsense mutation, an inversion, a single nucleotide polymorphism (SNP), a truncation, a point mutation, that changes the activity or expression of the one or more gene or nucleic acids. In some embodiments the change in expression is a reduction in expression or an elimination of the expression or activity, which expression can be the production of an encoded protein. The genetic modification can be made or be present in any sequence that affects expression or activity of the gene or nucleic acid sequence, or the nature or quantity of its product, for example to a coding or non-coding sequence, a promoter, a terminator, an exon, an intron, a 3' or 5' UTR, or other regulatory sequence. A genetic modification can be performed in any structure of the gene that results in attenuation or elimination of the gene or nucleic acid product or activity. In one embodiment the genetic modification is a deletion, disruption, or inactivation. The genetic modification can be made to, or be present in, the host cell's native genome. In some embodiments, a recombinant cell or organism having attenuated expression of a gene as disclosed herein can have one or more mutations, which can be one or more nucleobase changes and/or one or more nucleobase deletions and/or one or more nucleobase insertions, into the region of a gene 5' of the transcriptional start site, such as, in non-limiting examples, within about 2 kb, within about 1.5 kb, within about 1 kb, or within about 0.5 kb of the known or putative transcriptional start site, or within about 3 kb, within about 2.5 kb, within about 2 kb, within about 1.5 kb, within about 1 kb, or within about 0.5 kb of the translational start site. The genetic modification can comprise the deletion of one or more amino acids in a polypeptide sequence of a WD40 repeat containing protein.

A frameshift mutation is a particular type of mutation that involves either insertion or deletion of bases of DNA, where the number of inserted or deleted bases is not divisible by three. As a result a frameshift mutation disrupts the natural codon reading frame. This can result in insertion of an amino acid other than the naturally occurring programmed amino acid. Some or all of the entire DNA sequence following the mutation can also be disrupted or read incorrectly, i.e., as programming an amino acid other than the naturally programmed amino acid. In some embodiments a frame shift

mutation can result in premature termination of translation. Thus, in such embodiments the encoded polypeptide sequence can be much shorter, and unable to accomplish its natural function in the cell or organism. Thus, the gene or nucleic acid sequence has been functionally deleted. A nonsense mutation is the substitution of one or more base pair(s) that leads to a stop codon being inserted into a nucleic acid sequence. As a result, a nonsense mutation results in premature termination of translation, and can mean the functional deletion of the gene or nucleic acid sequence.

An “attenuation” is a genetic modification resulting in a reduction of the function, activity, or expression of a gene or nucleic acid sequence compared to a corresponding (control) cell or organism not having the genetic modification being examined, i.e., the diminished function, activity, or expression is due to the genetic modification. The activity of a nucleic acid sequence can be expression of an encoded polypeptide, a binding activity, the amount of signal transduction or transcription regulation, or other activity the nucleic acid sequence exerts within the organism. In various embodiments an attenuated gene or nucleic acid sequence disclosed herein produces less than 90%, or less than 80%, or less than 70%, or less than 50%, or less than 30%, or less than 20%, or less than 10%, or less than 5% or less than 1% of its function, activity, or expression of the gene or nucleic acid sequence compared to the corresponding (control) cell or organism. In various embodiments a gene attenuation can be achieved via a deletion, a disruption, or an inactivation. Any of the genetic modifications described herein can result in partial or complete attenuation of the function, activity, or expression of the attenuated gene or nucleic acid sequence, which in some embodiments can lead to a level of function, activity, or expression that is not substantially more than that of complete attenuation; for example, the function, activity, or expression can yield a result that is less than 10% different from a recombinant cell or organism having a complete attenuation of the gene or nucleic acid sequence.

An unmodified gene or nucleic acid sequence present naturally in the organism denotes a natural, endogenous, or wild type sequence. A deletion can mean that at least part of the object nucleic acid sequence is deleted or that at least part of the encoded product is eliminated. But a deletion can also be accomplished by disrupting a gene through, for example, the insertion of a sequence into the gene that is not naturally present (e.g., a selection marker), a combination of deletion and insertion, or mutagenesis resulting in insertion of a stop codon. But a deletion can also be performed by other genetic modifications known to those of ordinary skill that result in the loss of expression, activity, or function of a gene or nucleic acid sequence.

A functional deletion is a genetic modification that removes at least so much of the activity or expression of a gene or nucleic acid sequence so that any remaining activity or expression of the gene or nucleic acid sequence has no significant effect on the cell or organism compared to a corresponding (control) cell or organism not having the functional deletion and cultivated under the same or substantially the same conditions. In some embodiments the functional deletion can remove all function, activity, or expression of the gene or nucleic acid sequence. A functional deletion can involve an at least partial deletion of the coding or non-coding sequence of the gene. A “deletion” or “knock out” removes all function, activity, or expression of a gene or nucleic acid sequence. A “disruption” of a gene is a functional deletion by insertion or deletion of a nucleotide sequence into or from the coding, non-coding, or regulatory portion of a gene with resulting partial or complete loss of

function, activity, or expression of the gene. An “inactivation” is a type of functional deletion causing partial or complete loss of activity or expression of an inactivated gene or nucleic acid sequence. An “inactivation” can be reversible or irreversible (for example the reversible or irreversible binding of a component to the gene or nucleic acid sequence). Thus, deletions, functional deletions, inactivations, and disruptions can also be attenuations; disruptions and inactivations can also be functional, i.e., can remove at least so much of the activity or expression of a gene or nucleic acid sequence so that any remaining activity or expression of the gene or nucleic acid sequence has no significant effect on the cell or organism compared to a corresponding (control) cell or organism. An attenuation can also be a downregulation of a gene or nucleic acid sequence, which refers to the cell or organism decreasing the amount of function, activity, or expression. Functional expression refers to the expression of a functional product or activity of a nucleic acid sequence. When the expressed product of a nucleic acid is a polypeptide, functional expression means expression of polypeptide activity having at least 10% or at least 25% or at least 50% or at least 75% of the activity of a corresponding (control) cell or organism not having the modification and cultivated under the same or substantially the same conditions. For activity of a gene or nucleic acid sequence functional expression means activity or expression of at least 10% or at least 25% or at least 50% or at least 75% of the activity or expression of a corresponding (control) cell or organism not having the modification at issue and cultivated under the same or substantially the same conditions. Thus, various types of genetic modifications can be given terms that overlap in description. Persons of ordinary skill know that the particular term describing a genetic modification can be dependent both on how a gene or its components, or nucleic acid sequence is being physically changed as well as on the context. The recombinant cells or organisms of the invention can have any of the types of genetic modifications described herein.

In one embodiment the genetic modification is a deletion (or “knock out”) involving the introduction of a stop codon into a gene or nucleic acid sequence encoding a WD40 repeat containing protein described herein or variant thereof. For example the genetic modification can be a stop codon or frame shift mutation (or other genetic modification described herein) introduced into a gene or nucleic acid of any one of SEQ ID NO: 14-26 or a variant of any, or in a gene or nucleic acid sequence encoding any one of SEQ ID NOs: 1-13 (WD40 repeat containing proteins), or a variant of any. In one embodiment the genetic modification is a stop codon or frame shift mutation inserted into a gene or nucleic acid sequence of SEQ ID NO: 15 or 16 or a variant of either, or encoding SEQ ID NO: 1, or a variant thereof. In another embodiment the genetic modification is a stop codon or frame shift mutation inserted into a gene or nucleic acid sequence encoding SEQ ID NO: 8, or a variant thereof (WD40 repeat containing protein of *Parachlorella*). Variant sequences of any sequence described herein have at least 60% sequence identity or at least 70% sequence identity or at least 75% or at least 80% or at least 85% or at least 90% or at least 95% or at least 98% sequence identity to any nucleotide or polypeptide sequence of any of SEQ ID NOs: 1-26. In some embodiments a variant sequence can also have at least 25 or at least 50 or at least 75 or at least 100 or at least 150 or at least 200 or at least 250 or at least 300 or at least 350 or at least 400 contiguous amino acids of any one of SEQ ID NOs: 1-13, and optionally, also have a recited percent identity. In other embodiments the variants sequence

can have at least 500 or at least 600 or at least 700 or at least 800 or at least 1000 or at least 1200 or at least 1400 or at least 1500 contiguous nucleotides of any sequence of SEQ ID NO: 14-26 and, optionally, also have a recited percent identity.

In one embodiment the genetic modification is a modification that results in a stop codon, nonsense mutation, or frame shift mutation (or other genetic modification described herein) at the sequence coding for SEQ ID NO: 14 (AGACTCGCACCG) (V204fs), or a variant thereof, or in SEQ ID NO: 15 or 16, or a variant of either. The genetic modification can also be targeted to a regulatory sequence with the effect of eliminating or diminishing the activity or expression of a gene or nucleic acid sequence of SEQ ID NO: 14-26 or a variant of any, or a gene or nucleic acid sequence that encodes for any one SEQ ID NO: 1-13 or a variant of any of them.

In some embodiments the genetic modification can also be a frame shift mutation, a stop codon mutation, or a nonsense mutation introduced into a gene or nucleic acid sequence (including regulatory sequences supporting such gene or nucleic acid sequences) encoding any WD40 repeat containing protein disclosed herein. In various embodiments the gene or nucleic acid sequence is of SEQ ID NO: 14-26, or a variant of any; in a specific embodiment the gene or nucleic acid sequence encodes for a polypeptide of SEQ ID NO: 1 (or a variant thereof) or a polypeptide of SEQ ID NO: 8 (or a variant thereof), which mutation can be introduced at any location of the sequence or into a regulatory sequence governing the sequence, where the modification results in a termination of transcription from the gene prior to its natural point. Thus, in one embodiment the mutation is the introduction of a frame shift mutation, stop codon, or nonsense mutation that functionally deletes or disrupts the activity or expression of the gene or nucleic acid sequence. The frame shift mutation, stop codon, or nonsense mutation (or other modification) can also be introduced at any of many different loci or locations within a gene encoding a WD40 repeat containing protein or in a regulatory sequence, for example at a promoter, terminator or other regulatory sequence, that attenuates the gene or the activity of the encoded polypeptide, and that results in functional deletion of the gene. Analogous modifications can be made to the sequence(s) for similar effect. Such insertion or deletion or other mutation can also cause a loss of expression, function or activity in the encoded WD40 repeat containing protein, and result in the effect of increased lipid productivity.

In other embodiments the genetic modification can be to a nucleic acid sequence of SEQ ID NO: 15 or 16, or to a variant of either having at least 70% sequence identity or at least 75% or at least 80% or at least 85% or at least 90% or at least 95% or at least 98% sequence identity to any nucleotide sequence of SEQ ID NOs: 15 or 16.

Any of the recombinant cells or organisms of the invention can have a reduced functional absorption cross section of PSII and/or reduced PSII antenna size. For example, the cross-sectional unit size of the PSII antenna can be reduced by at least about 10%, at least 20%, at least 30%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least about 70%, or at least about 80% compared to the functional absorption cross section of PSII and/or PSII antenna size of the corresponding (control) cell or organism not having the genetic modification. In some embodiments the recombinant cells or organisms of the invention can additionally (and optionally) have a reduced functional absorption cross section of PSI or reduced PSI antenna size by the same amounts stated above

versus a corresponding (control) cell or organism. All statements herein of increases in biochemical parameters (e.g., FAME productivity or TOC productivity) are versus a corresponding (control) cell or organism unless indicated otherwise.

In some embodiments, a recombinant algal cell or organism as provided herein can have increased Fv/Fm with respect to a corresponding control cell or organism. For example, the mutant photosynthetic organism may have Fv/Fm increased by at least 5%, at least 10%, at least 12%, at least 15%, at least 20%, at least 30%, at least 40% or at least 50% compared to a corresponding (control) photosynthetic organism. In various embodiments the Fv/Fm can be increased by 5-50%, or by 5-30% or by 5-20% with respect to a control photosynthetic organism.

Further, a mutant photosynthetic organism as provided herein can have an increased rate of electron transport on the acceptor side of photosystem II with respect to a control or wild type cell. The rate can be at least about 20%, 30%, 40%, 50%, 60%, 80%, or 100% higher compared to a corresponding control cell or organism. In addition, mutant photosynthetic cells or organisms of the invention can have a rate of carbon fixation (P_{max} (C)) in a recombinant cell or organism as provided herein can be elevated with respect to a control organism. For example, P_{max} (14 C) can be increased by at least about 20%, 30%, 40%, 50%, 60%, 80%, or 100% compared to a corresponding control cell or organism.

In some embodiments, the recombinant cells or organisms of the invention have decreased PSI and/or PSII antenna size and can optionally also have a higher amount of a ribulose biphosphate carboxylase activase (Rubisco activase or "RA") than a corresponding (control) or wild type organism, for example, at least 1.2, 1.4, 1.6, 1.8, 2, 2.2, or 2.5 fold the amount of RA as a control organism. In some embodiments, the mutants demonstrate reduced expression of 6, 8, 10, 12, or 14 LHCP genes and increased expression of an RA gene, such as an RA-a or RA-P gene. Thus, the recombinant cells or organisms of the invention can be mutant photosynthetic organisms having reduced chlorophyll and reduced PSII antenna size where the mutants have a higher amount of Rubisco activase than control photosynthetic organisms.

The LHC super-gene family encodes the light-harvesting chlorophyll a/b-binding (LHC) proteins that constitute the antenna system of the photosynthetic apparatus. A recombinant algal mutant of the invention can also have a reduced expression of one or more LHC genes. Thus, in some embodiments the recombinant cells or organisms of the invention have at least 6, at least 8, at least 10, or at least 12 LHC genes that are attenuated or downregulated with respect to their expression level in a corresponding (control) cell or organism. In various embodiments the reduction in expression of the one or more LHC genes can be a reduction of at least 20%, or at least 30%, or at least 40%, or at least 50%, or at least 60%, or at least 70% in the level of LHC transcripts compared to the control cell or organism.

The structure of a gene consists of many elements, of which the protein coding sequence is only one part. The gene includes nucleic acid sequences that are not transcribed and sequences that are untranslated regions of the RNA. Genes also contain regulatory sequences, which includes promoters, terminators, enhancers, silencers, introns, 3' and 5' UTRs, and coding sequences, as well as other sequences known to be a part of genes. In various embodiments any of these structures or nucleic acid sequences (e.g., any of SEQ ID NO: 14-26 or variants of any) can have one or more of

the genetic modifications described herein that result in the higher lipid productivity and/or higher biomass productivity as described herein.

WD40 Repeat Containing Protein

WD40 repeat containing proteins (or domains) have four or more repeating units containing a conserved core motif that ends with tryptophan-aspartic acid (WD). Proteins of this class have domain folds that form a beta propeller structure, which are symmetric folds having 4-9 anti-parallel, four-stranded beta sheets arranged radially around a central axis. The domains form a platform on which protein complexes assembly reversibly. They are believed to play a key role in the formation of protein-protein complexes. In any embodiment the genetic modification to the WD40 repeat containing protein can be to any WD40 repeat domain of the protein.

The recombinant algal cell or organism of the invention can have a genetic modification described herein to a nucleic acid sequence of any of SEQ ID NO: 14-26, or to a nucleic acid sequence having at least 60% sequence identity or at least 70% sequence identity or at least 75% or at least 80% or at least 85% or at least 90% or at least 95% or at least 98% sequence identity to any of SEQ ID NOS: 14-26. The nucleic acid sequence can encode a WD40 repeat containing protein. The genetic modification can also be to any of the disclosed sequences and also have at least 500 or at least 600 or at least 700 or at least 800 or at least 1000 or at least 1200 or at least 1400 or at least 1500 contiguous nucleotides of any sequence of SEQ ID NO: 14-26 (and, optionally, also have a recited percent identity).

The genetic modification can be to a gene or nucleic acid sequence that encodes a WD40 repeat containing protein having a polypeptide sequence of any one of SEQ ID NO: 1-13 or a variant of any of said nucleic acid sequences, and the genetic modification can result in the higher lipid productivity and/or higher biomass productivity as described herein. In various embodiments the genetic modification can be to a nucleic acid sequence that encodes a WD40 repeat containing protein (or domain) having at least 60% sequence identity or at least 70% sequence identity or at least 75% or at least 80% or at least 85% or at least 90% or at least 95% or at least 98% sequence identity to any of SEQ ID NOS: 1-13. In some embodiments the nucleic acid sequence can be a variant as stated and also encode at least 25 or at least 50 or at least 75 or at least 100 or at least 150 or at least 200 or at least 250 or at least 300 or at least 350 or at least 400 contiguous amino acids of any one of SEQ ID NOS: 1-13, or a variant of any.

In any embodiment the WD40 repeat containing protein can be a functional WD40 repeat containing protein. In various embodiments the recombinant algal cell or organism can have a genetic modification described herein to a nucleic acid sequence of SEQ ID NO: 15 or 16, or to a variant of either having at least 70%, or at least 75%, or at least 80%, or at least 85%, or at least 90%, or at least 95%, or at least 98% sequence identity.

Persons of ordinary skill know how to calculate the percent of "sequence identity" between two sequences. Any method of determining sequence identity that has acceptance by most persons of ordinary skill in the art or otherwise widely accepted in the field can be utilized to determine the sequence identity between two sequences. In one embodiment the percent of sequence identity can be determined by BLAST (Basic Local Alignment Search Tool) analysis using the algorithm employed by the programs blastp, blastn, blastx, tblastn, and tblastx (Altschul (1997), Nucleic Acids Res. 25, 3389-3402, and Karlin (1990), Proc. Natl. Acad.

Sci. USA 87, 2264-2268). In one embodiment the search parameters for histogram, descriptions, alignments, expect (i.e., the statistical significance threshold for reporting matches against database sequences), cutoff, matrix, and filter (low complexity) can be at the default settings. The default scoring matrix used by blastp, blastx, tblastn, and tblastx can be the BLOSUM62 matrix (Henikoff (1992), Proc. Natl. Acad. Sci. USA 89, 10915-10919). For blastn the scoring matrix can be set by the ratios of M (i.e., the reward score for a pair of matching residues) to N (i.e., the penalty score for mismatching residues), wherein the default values for M and N can be +5 and -4, respectively. Four blastn parameters can be adjusted as follows: Q=10 (gap creation penalty); R=10 (gap extension penalty); wink=1 (generates word hits at every winkth position along the query); and gapw=16 (sets the window width within which gapped alignments are generated). The equivalent Blastp parameter settings for comparison of amino acid sequences can be: Q=9; R=2; wink=1; and gapw=32. A Bestfit comparison between sequences, available in the GCG package version 10.0, can use DNA parameters GAP=50 (gap creation penalty) and LEN=3 (gap extension penalty), and the equivalent settings in protein comparisons can be GAP=8 and LEN=2.

Laboratory Strains

In various embodiments the one or more genetic modification(s) can be made in (i.e., derived from) a cell or organism that is a wild type, parent, or laboratory strain. Laboratory strains are cells or organisms that have been cultured in a laboratory setting. As a result of culturing over a period of time the strain has undergone some adaptation(s) advantageous to growth in the laboratory setting. A laboratory strain therefore can develop one or more advantageous characteristics from adaptations deriving from time of culturing in the laboratory environment. For example, laboratory strains can have higher biomass productivity and/or higher lipid productivity than a wild-type strain. In some embodiments one or more genetic modifications disclosed herein can be performed on a laboratory strain to result in a recombinant algal cell or organism of the invention. In such embodiments the laboratory strain can therefore be a corresponding control algal cell or organism described herein that does not have the genetic modification being considered. A test cell can be derived from the corresponding control cell or organism. For example, the corresponding control cell or organism can be a parental strain of the test cell or organism.

Increased Lipid Productivity

The recombinant mutant algae of the invention having a genetic modification to a gene or nucleic acid sequence encoding a WD40 repeat containing protein or other modification described herein can demonstrate an increase in the production of lipid in the cell or organism versus a corresponding (control) cell or organism. The increase in lipid production can be measured by any accepted and suitable method, for example using fatty acid methyl ester (FAME) analysis. In one embodiment the increase in lipid production is measured as an increase in total FAME produced by the recombinant organisms. The recombinant cells or organisms of the invention have a genetic modification to a gene or nucleic acid sequence encoding a WD repeat containing protein or domain and can exhibit at least 20% or at least 25% or at least 30% or at least 40% or at least 50% or at least 60% or at least 70% or at least 80% or at least 90% or at least 100% greater lipid productivity compared to a corresponding control cell or organism, as described herein. In other embodiments the increase in lipid productivity can be 15-25%, or 15-35%, or 15-45%, or 15-50%, or 20-25%, or 25-45%, or 25-55%, or 25-70%, or 25-90%, or 25-100%, or

25-150% or 25-200% or 30-35% or 30-45% or 30-55%. The increase can be weight for weight (w/w) of lipid. In any embodiment the lipid can be fatty acid. In one embodiment lipid productivity is measured using the FAME profile (fatty acid methyl ester) of the respective cells or organisms. In one embodiment lipid productivity can be expressed as mg/L. In other embodiments the recombinant cells or organisms of the invention can exhibit at least 50 g/m² or at least 60 or at least 70 or at least 80 grams per square meter of FAME accumulation after 5 days of cultivation. Methods of producing a FAME profile are known to persons of ordinary skill in the art. A FAME profile can be determined using any suitable and accepted method, for example a method accepted by most persons of ordinary skill in the art. The recombinant cell or organisms of the invention can, optionally, also have an increase in biomass productivity can be 15-35% or 15-40% or 25-45% or 15-50% or 25-70% or 50-100% or 50-200% (w/w).

An increase in lipid production or lipid productivity can be measured by weight, but can also be measured in grams per square meter per day of the surface of a cultivation vessel (e.g., a flask, photobioreactor, cultivation pond). In various embodiments the recombinant cell or alga of the invention produces at least 3 or at least 4 or at least 5 or at least 6 or at least 7 or at least 8 or at least 10 or at least 12 or at least 13 or at least 14 grams per square meter per day of lipid production, which can be measured by the FAME profile. In any of the embodiments the high lipid and/or high biomass productivity phenotype can be obtained under nitrogen deplete conditions, which in some embodiments can involve dilution and/or replacement of medium with fresh nitrogen deplete medium during growth. Dilutions can be by any suitable amount, for example dilution by about 50% or by about 60% or by about 70% or at least 70%, or by about 80%, or by more than 80%, which can be done every day, every 2 days, every 3 days or every 4 days or every 5 days. In one embodiment the lipid product is a fatty acid and/or a derivative of a fatty acid. In one embodiment the fatty acids and/or derivatives of fatty acid comprise one or more species of molecules having a carbon chain between C8-C12, and/or C8-C18 and/or C8-C20 and/or C8-C22 and/or C8-C24, hereby disclosed in all possible combinations and sub-combinations. In one embodiment the growth conditions can be batch growth, involving spinning cells to remove nitrogen from the medium, replacing with nitrogen deplete medium, and resuming batch growth.

In any of the embodiments the genetic modification to the gene or nucleic acid sequence encoding a WD repeat containing protein can result in an attenuation of expression of the respective genes. The genetic modification of any one or more of these genes or nucleic acids can be any of those described herein, e.g., any one or more of SEQ ID NOs: 14-26. In one embodiment the genetic modification is a deletion, disruption, or inactivation. In another embodiment the genetic modification is a deletion (which optionally, can be a functional deletion) or a disruption of the gene.

Biomass Productivity

The recombinant algal cells of the invention having a genetic modification to a gene or nucleic acid sequence encoding a WD40 repeat containing protein described herein, and can also have higher biomass productivity than a corresponding (control) organism not having the genetic modification. Biomass can be measured as an increase in the total organic carbon (TOC) produced by the cell or organism, for example measured by TOC analysis known to persons of ordinary skill in the art. The recombinant cells can have at least 12% higher, or at least 15% higher, or at

least 20% higher or at least 25% higher or at least 30% higher or at least 35% higher, or at least 50% higher or at least 60% higher or at least 70% higher or at least 80% higher or at least 90% higher or at least 100% higher or at least 125% higher or at least 150% higher biomass productivity than a corresponding (control) cell or organism, which can be measured by total organic carbon analysis. In other embodiments the biomass productivity can be 10-15% higher or 15-25% higher or 20-25% higher or 20-30% higher or 20-40% higher or 20-45% higher. The increase in biomass productivity can, optionally, be in addition to the increase in lipid productivity.

Various methods of measuring total organic carbon are known to persons of ordinary skill in the art. Biomass productivity can be measured as mg/ml of culture per time period (e.g., 1 day or 2 days or 3 days or 4 days or 5 days). In some embodiments the higher biomass productivity and/or higher lipid productivity as described herein can occur under nitrogen deplete conditions. Thus, in one embodiment the recombinant alga or cell of the invention can have higher lipid production and/or higher total organic carbon production than a corresponding (control) cell or organism, which higher amount can be produced under nitrogen deplete or low nitrogen conditions. Nitrogen deplete conditions can involve culturing in a buffer having less than 0.5 mM of nitrogen in any available form external to the cell or organism. In some embodiments the cells can be cultured in 0.5 mM or less of KNO₃ or urea as a nitrogen source. Other buffers may also be used and be nitrogen deplete [[if they contain a level of nitrogen that does not change the physiology of a nitrogen-related parameter (e.g., lipid productivity or biomass productivity) by more than 10% versus culturing the cell in a medium free of a nitrogen source external to the cells or organisms.]] In any embodiment biomass productivity can be evaluated by measuring an increase in the total organic carbon of the cells. Nutrient replete conditions are those where the growth of the cultivated organism is not limited by a lack of any nutrient. A nitrogen free medium is also considered nitrogen deplete conditions.

Methods of Producing Lipid

The invention also provides methods for producing a lipid containing product. The methods involve culturing a recombinant algal cell or organism described herein to thereby produce a lipid product. Any of the methods can also involve a step of harvesting lipid produced by the recombinant algal cell or organism described herein. The culturing can be for a suitable period of time, for example, at least 1 day or at least 3 days or at least 5 days.

The invention also provides methods for producing a composition containing lipids. In one embodiment the methods can involve performing a genetic modification to an algal organism in a gene or nucleic acid sequence encoding a WD40 repeat containing protein. The methods can also involve a step of culturing a recombinant algal cell or organism described herein to thereby produce a composition containing lipids. The composition can be a biomass composition or biomass product containing any one or more recombinant algae described herein. The cultivating can be done in any suitable medium conducive to algal growth (e.g., an algal growth medium or any medium described herein). The methods can also involve a step of harvesting lipids from the composition or biomass containing lipids. The methods can involve a step of harvesting lipids from the recombinant cells or organisms. Any of the methods herein can also involve a step of purifying the lipid containing composition to produce a biofuel or biofuel precursor. A

biofuel precursor is a composition containing lipid molecules that can be purified or otherwise converted into a biofuel.

The invention also provides methods of producing a recombinant algal cell or organism having higher lipid productivity and/or higher biomass productivity than a corresponding control cell or organism. The methods involve exposing algal cells or organisms to ultraviolet light to produce a recombinant cell or organism described herein that has higher lipid productivity than a corresponding control cell or organism. In one embodiment algal organisms having higher lipid productivity can be identified by contacting the recombinant cells with a stain that identifies lipids (e.g., by BODIPY dye). Optionally methods can include a step of isolating lipids from the recombinant algal organisms. Organisms having higher biomass productivity can be identified by TOC analysis. The recombinant alga can be cultivated in any suitable growth media for algae, such as any of those described herein. The uv treatment can involve, for example, subjecting the culture to uv light, or gamma radiation, or both, for a suitable period of time or under a suitable uv regimen or gamma radiation regimen. Persons of ordinary skill understand suitable regimens for uv exposure for mutagenesis. The uv regimen can involve exposing the cells or organisms to uv radiation, which can be performed in batches with each batch receiving a dose. Multiple cell batches can receive different doses of energy for each batch of cells. Subjecting the algal organism to ultra-violet light can comprise exposing the cell or organism to at least 10 or at least 15 uJ/cm² of energy, which exposure can be performed within 1 second or within 5 seconds or within 30 seconds or within 1 minute. In some embodiments 4 or 5 batches of cells can receive doses of exposure to 16-57 uJ/cm² of energy, and exposure energy can increase with each separate batch. The cell batches can be pooled together after exposures are complete. The recombinant alga (or pooled algae) can be cultivated for at least 2 days or at least 3 days, or at least 4 days, or at least 5 days, or at least 6 days, or at least 10 days, or at least 20 days, or from 2-10 days, or from 2-20 days or from 2-25 days after exposure. The recombinant algal organisms can be any described herein. The uv regimen can involve exposing the cells or organisms to multiple exposures to uv radiation, which can become progressively stronger.

Any of the recombinant cells or organisms of the invention can be cultivated in batch, semi-continuous, or continuous culture to produce the higher biomass productivity and/or higher lipid productivity. In some embodiments the culture medium can be nutrient replete, or nitrogen deplete (—N). In some embodiment the culturing is under photoautotrophic conditions, and under these conditions inorganic carbon (e.g., carbon dioxide or carbonate) can be the sole or substantially the sole carbon source in the culture medium.

The invention also provides a biofuel comprising a lipid product of any of the recombinant cells or organisms described herein. The biofuel is produced by purifying a lipid containing composition produced by a recombinant algal cell or organism described herein. The biofuel can be produced by purifying or otherwise converting a lipid product produced by a recombinant algal cell or organism described herein to produce a biofuel. In one embodiment the biofuel is a biodiesel fuel, which can contain fatty acid methyl ester molecules in sufficient proportion to be combustible in an internal combustion engine. In various embodiments the biofuel can contain carbon chains with an average carbon chain length of at least 5 carbon atoms, or at least 8 carbon atoms, or at least 10 carbon atoms.

FAME and TOC Analysis Methods

The lipid productivity of the cells or organisms can be measured by any method accepted in the art, for example as an increase or decrease in fatty acid methyl esters comprised in the cell, i.e., FAME analysis. In various embodiments any of the recombinant algal cells or organisms of the invention can have higher biomass productivity as described herein versus corresponding control cells or organisms. In some embodiments the recombinant algal cells or organisms of the invention can have higher lipid productivity and also higher biomass productivity compared to a corresponding control cell or organism. Biomass productivity can be measured by any methods accepted in the art, for example by measuring the total organic carbon (TOC) content of a cell. Embodiments of both methods are provided in the Examples.

“FAME lipids” or “FAME” refers to lipids having acyl moieties that can be derivatized to fatty acid methyl esters, such as, for example, monoacylglycerides, diacylglycerides, triacylglycerides, wax esters, and membrane lipids such as phospholipids, galactolipids, etc. In some embodiments lipid productivity is assessed as FAME productivity in milligrams per liter (mg/L), and for algae, may be reported as grams per square meter per day (g/m²/day). In semi-continuous assays, mg/L values are converted to g/m²/day by taking into account the area of incident irradiance (the SCPA flask rack aperture of 1½ inches×3¾", or 0.003145 m²) and the volume of the culture (550 ml). To obtain productivity values in g/m²/day, mg/L values are multiplied by the daily dilution rate (30%) and a conversion factor of 0.175. Where lipid or subcategories thereof (for example, TAG or FAME) are referred to as a percentage, the percentage is a weight percent unless indicated otherwise. The term “fatty acid product” includes free fatty acids, mono-di, or tri-glycerides, fatty aldehydes, fatty alcohols, fatty acid esters (including, but not limited to, wax esters); and hydrocarbons, including, but not limited to, alkanes and alkenes).

Embodiments

In one embodiment the invention provides a recombinant algal organism of the Class Trebouxiophyceae having a genetic modification in a gene or nucleic acid sequence encoding a WD40 repeat containing protein. The recombinant alga exhibits higher lipid productivity and/or biomass productivity versus a corresponding control algal cell not having the genetic modification. In various embodiment the Trebouxiophyceae organism can be from the family Oocystaceae or Chlorellaceae. In one embodiment the organism is of the genus *Oocystis*.

In one embodiment the invention provides a recombinant Trebouxiophyceae organism having a deletion, disruption, or inactivation in a gene or nucleic acid sequence encoding a WD40 repeat containing protein. The deletion can be a functional deletion. In one embodiment the deletion, disruption, or inactivation involves the insertion of a nonsense mutation or a frameshift mutation in a gene or nucleic acid sequence encoding a WD40 repeat containing protein. In one embodiment the encoded WD40 repeat containing protein can have at least 80% or at least 85% or at least 90% or at least 95% sequence identity to SEQ ID NO: 1 or 8. In another embodiment the recombinant cell or organism can have the deletion, disruption, or inactivation in a nucleic acid sequence encoding a WD40 repeat containing protein, which be a nucleic acid having at least 80% or at least 90% or at least 95% sequence identity to SEQ ID NO: 15 or SEQ ID NO: 16. The recombinant alga exhibits higher lipid productivity and/or biomass productivity versus a corre-

sponding control algal cell not having the genetic modification. The alga can be a Trebouxiophyceae organism, for example from the family Oocystaceae, and optionally from the genus *Oocystis*. The increase in lipid productivity can be an increase in 2-day FAME accumulation of at least 20% w/w, or at least 25%, or 30-50% or 30-55%, which can, optionally, be cultivated under nitrogen deplete conditions. The recombinant cells or organisms can, optionally, also have an increase in biomass productivity of at least 12% or at least 15% or at least 20% or 15-25% or 20-40%, which can be expressed as 2-day TOC accumulation. Thus in one embodiment the recombinant cells or organisms have an increase in lipid productivity (as measured by FAME profile) of 30-55% and an increase in biomass productivity of at least 20% versus a control cell or organism. In another embodiment the increase in lipid productivity can be 20-40%. The increase in FAME accumulation and/or TOC accumulation can be measured after two days or after five days cultivation, and can be in nitrogen deplete conditions. The recombinant cells or organisms can also have a FAME/TOC ratio of greater than 0.3 after 2 days of cultivation.

In another embodiment the cells or organisms of the invention can have an increase in lipid productivity of at least 20% (as measured by 2-day FAME accumulation) and, optionally, an increase in biomass productivity of at least 12% (as measured by TOC); and optionally also have a FAME/TOC ratio of greater than 0.3 or 0.4. In another embodiment the cells or organisms of the invention can have an increase in lipid productivity of at least 20% (as measured by 2-day FAME accumulation); and optionally also have a FAME/TOC ratio of greater than 0.5.

In one embodiment the invention provides a recombinant algal organism of the Class Trebouxiophyceae having a genetic modification to a gene or nucleic acid sequence encoding a WD40 repeat containing protein. In one embodiment the gene or nucleic acid sequence is any one of SEQ ID NO: 14-26 or a variant of any, or can be a gene or nucleic acid sequence encoding any one of SEQ ID NO: 1-13, or a variant of any. The genetic modification can be a deletion (optionally a functional deletion) or disruption of the gene or nucleic acid sequence. The deletion can be a functional deletion. The recombinant alga exhibits higher lipid productivity and, optionally, higher biomass productivity versus a corresponding control algal cell not having the genetic modification. In various embodiment the Trebouxiophyceae organism can be from the family Oocystaceae or Chlorellaceae. In one embodiment the organism is of the genus *Oocystis*. The increase in lipid productivity can be an increase in 2-day or 5-day FAME accumulation of at least 20% w/w, or 30-50% or 30-55%, which can, optionally, be cultivated under nitrogen deplete conditions. The recombinant cells or organisms can, optionally, also have an increase in biomass productivity of at least 15% or at least 20% or 15-25% or 20-40%, which can be expressed as 2-day or 5-day TOC accumulation. Thus in one embodiment the recombinant cells or organisms have an increase in lipid productivity of 30-55% and an increase in biomass productivity of at least 20%. In another embodiment the increase in lipid productivity can be 20-40%. The increase in FAME accumulation and/or TOC accumulation can be measured after two days or after five days cultivation in nitrogen deplete conditions. In another embodiment the cells or organisms of the invention can have an increase in lipid productivity of at least 50% (as measured by 5-day FAME accumulation); and optionally also have a FAME/TOC ratio of greater than 0.5 or 0.6 after 5 days cultivation.

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In all embodiments the increase in lipid productivity and/or TOC productivity can be relative to a corresponding (control) cell or organism, which can be a wild-type cell or a laboratory strain.

EXAMPLE 1

Various aliquots of 10 mL of Strain 15 (STR00015 or “wild type”) *Oocystis* sp. were acclimated to diel growth conditions on urea supplemented minimal media. Different aliquots of cells at a concentration of 2×10^6 cells/mL were then exposed to different amounts of UV radiation (22.4, 33.6, 44.8 and 56 mJ/cm² using at total of 50, 50, 50 and 20 mL of culture, respectively) in a UV crosslinker apparatus. The mutagenized cell aliquots were allowed to recover in the dark for 48 hrs. Cultures were then scaled up in low light (~100 uE) before a first round of enrichment. The mutagenized cells were then acclimated to diel growth at a light intensity of ~100 uE and 1% CO₂ in urea supplemented minimal medium for a week. The cultures were scaled up for 3 days, bubbled with 1% CO₂ at a maximum irradiance of ~1400 uE under diel conditions, to an OD730 of about 1.0. The cultures was then centrifuged at 5000 g for 10 mins and the cell pellets resuspended in nitrogen-free minimal medium to an OD730 of about 0.9. This nitrogen-free culture was then incubated for 48 hrs in square-bottom flasks bubbled with 1% CO₂ at a maximum irradiance of about 1400 uE under diel conditions. After 48 hours in nitrogen-free batch growth, an aliquot of cells was removed and subjected to staining with the lipid specific dye BODIPY for 10 mins in the dark at a final concentration of 0.2 ug/mL.

Mutant cells having the highest level of BODIPY staining were enriched by fluorescence activated cell sorting (FACS). Enriched cell populations were scaled up and starved for nitrogen as described above, and then subjected to additional rounds of BODIPY-based FACS enrichment. This iterative

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process was repeated for a total of five rounds, with the top 2%, 1%, 1%, 0.25% and 0.25% of the population retained in each iteration.

The final sort was plated on minimal medium agar plates supplemented with urea to isolate single axenic colonies.

Isolates were scaled up in T25 tissue culture flasks in minimal medium supplemented with urea, then transitioned to nitrogen-free minimal medium for 48 hrs. The lipid and biomass accumulation of isolated mutants were compared to the parental strain STR00015, with lipid content measured by total fatty acid methyl ester (FAME) analysis and biomass measured by total organic carbon (TOC). As can be seen in FIG. 1, several isolates from the screen showed an increase in accumulated FAME and TOC, as well as FAME/TOC—an indicator of how much fixed carbon is partitioned to lipids. This indicated that mutants with improved lipid productivity had been isolated by the process described. Strain 27434 was identified as having high FAME and biomass accumulation.

EXAMPLE 2

Genomic DNA was isolated from Strain 27434 along with parental (wild type) Strain 15 and sequenced on high throughput sequencing system generating 150 bp paired end reads. Reads were processed, mapped to the wild type strain as reference genome and analyzed by a small variants algorithm. An example of a small variants algorithm is the Freebayes polymorphism detection software, although other programs can also be successfully utilized. Analysis of SNPs and small insertions/deletions (InDels) revealed Strain '434 contained a total of 170 polymorphisms. Twenty-four of these mutations were located within exons or at splice junctions and were prioritized for Cas9-mediated gene deletion as they had the highest probability of altering gene function and/or activity, as shown in Table 1 below.

TABLE 1

Mutations identified within exons or at splice junctions in Strain '434					
Transcripts	Descriptions	Type	Ref	Alt	AAMod
1 3EUKT2015873	Protein kinase-like (PK-like)	SNP	G	A	Gln1301*
2 3EUKT2027500	YVTN repeat-like/Quinoprotein amine dehydrogenase	Deletion	AGACTC GCACCG	AG	Val204fs
3 3EUKT2031181	CAAX amino terminal protease self-immunity	SNP	G	A	Gln48*
4 3EUKT2028953	Proteasome subunit alpha type-1-A	SNP	C	T	Leu546Phe
5 3EUKT2014620	3-methylcrotonyl-CoA carboxylase alpha subunit	SNP	C	A	Ala426Ser
6 3EUKT2012690	DNA-directed DNA polymerase family protein	SNP	G	A	Ala627Val
7 3EUKT2036647	Conserved predicted protein	SNP	C	T	Glu640Lys
8 3EUKT2016841	Conserved predicted protein	SNP	T	A	Leu332Gln
9 3EUKT2025932	Chitin binding Peritrophin-A domain	SNP	C	T	Pro239Leu
10 3EUKT3238737	Retrovirus-related Pol polyprotein from transposon 17.6	SNP	G	A	Ser895Phe
11 3EUKT2017815	Phosphoglycerate mutase-like protein	Complex	CC	TT	Gly235Glu
12 3EUKT2020372	Conserved predicted protein	SNP	A	G	Thr118Ala
13 3EUKT2011248	RAP domain	SNP	G	A	Gly1722Asp
14 3EUKT2015114	Conserved predicted protein	SNP	G	A	Arg576His
15 3EUKT2013326	Manganese-dependent ADP-ribose/CDP-alcohol diphosphatase	SNP	G	A	Ser174Phe
16 3EUKT2015233	Phosphatidylinositol (PI) phosphodiesterase	SNP	C	T	Ser106Phe
17 3EUKT2015577	T-complex 1 subunit beta	SNP	G	A	Glu265Lys
18 3EUKT2015261	Cell surface protein homologous to bacterial outer membrane proteins	SNP	C	T	Pro171Ser
19 3EUKT2026772	Conserved predicted protein	SNP	G	A	Pro18Ser
20 3EUKT2021194	Conserved predicted protein	SNP	C	T	Pro48Leu
21 3EUKT2021524	Conserved predicted protein	SNP	C	T	Ser19Leu

TABLE 1-continued

Mutations identified within exons or at splice junctions in Strain '434					
Transcripts	Descriptions	Type	Ref	Alt	AAMod
22 3EUKT2018868	Conserved predicted protein	SNP	T	A	Asn84Ile
23 3EUKT2022105	Axonemal inner arm dynein heavy chain 4	SNP	G	A	Ala202Val
24 3EUKT2018529	Ribonuclease Inhibitor	SNP	G	A	Splice junction

The remaining 146 mutations were either intergenic or present in introns of a gene. An assessment of transcriptomics data from Strain 15 (wt) and Strain 27434 indicated that none of these 146 mutations had any significant impact on gene expression or transcript splicing.

EXAMPLE 3

Identification of Mutations

To identify which mutation(s) caused the high lipid phenotype in Strain 27434, independent knockouts of genes bearing SNPs in the strain (Table 1) were conducted via RNP-based Cas9-mediated gene disruption in two background strains: (i) the Strain 15 wild type parental strain; and (ii) Strain 24194—a laboratory strain evolved from Strain 15 and having improved biomass and lipid productivity.

All the strains generated were tested for improved biomass and lipid accumulation during nitrogen starvation in T25 flasks. From this analysis three independent lines were identified having a deletion of 3EUKT2027500, a gene encoding a WD40 repeat containing protein. These lines showed significantly higher biomass and lipid accumulation compared to the parental lines. One of these was constructed in the Strain 15 (wt) control strain (FIG. 2), while the other two were constructed in the Strain 24194 control strain (FIG. 3). As shown in FIG. 2, the first genetically engineered strain (identified as Strain 29857) showed a 22% and 15% improvement in accumulated FAME and TOC, respectively, compared to Strain 15. The strains engineered in the Strain 24194 control (identified as Strains 30525 and 30526) showed a 22% increase in FAME accumulation (FIG. 3).

EXAMPLE 4

Large scale productivity testing in 500 mL square-bottom flasks was conducted as described above, but over a 5 day period. The study revealed that the two knockout lines in laboratory Strain 24194 (Strain 30525 and 30526) showed a 14% and 21% improvement in lipid productivity, respectively (FIG. 4), confirming the results in the T25 flasks. The measured FAME/TOC was also substantially higher on each day of the experiment for both engineered strains (FIG. 4). The data therefore showed that deletion of 3EUKT2027500 provides a substantial increase in carbon partitioning and lipid productivity.

EXAMPLE 5

The amino acid sequence of 3EUKT2027500 (#2 in Table 1) was analyzed for functional domains and orthologs in other species. 3EUKT2027500 encodes a WD40 repeat containing protein with four tandem domains with weak similarity to WD40 repeats. WD40 repeats are short minimally conserved structural motifs of about 40 amino acids

often beginning with a glycine-histidine (GH) dipeptide and ending with tryptophan-aspartic acid (WD) dipeptide. Multiple copies of these repeats fold together to form the WD40 domain, which functions as a scaffold enabling interaction with proteins and nucleic acid and facilitating the formation of multiprotein complexes. WD40 repeat containing proteins are involved in a variety of cellular processes, including cell cycle progression, transcriptional regulation, signal transduction, apoptosis, biosynthesis of plant cell walls, anthocyanin biosynthesis, etc.

Further sequence identity analysis summarized in Table 2 revealed orthologs of 3EUKT2027500 are highly conserved and broadly distributed in green algae and plants.

TABLE 2

Alignment of Entire Protein Sequences of 3EUKT2027500 Orthologs				
	Organisms	Gene/Seq ID	% Identity	% Similarity [Positives]
1	<i>Coccomyxa subellipsoidea</i>	XP_005642574.1 SEQ ID NO: 2	78	89
2	<i>Chlamydomonas reinhardtii</i>	Cre16.g674000 SEQ ID NO: 3	69	78
3	<i>Volvox carteri</i>	XP_002950453.1 SEQ ID NO: 4	72	80
4	<i>Auxenochlorella protothecoides</i>	RMZ56784.1 SEQ ID NO: 5	62	76
5	<i>Chlorella sorokiniana</i>	PRW57951.1 SEQ ID NO: 6	70	81
6	<i>Chlorella variabilis</i>	XP_005847572.1 SEQ ID NO: 7	69	80
7	<i>Parachlorella</i> sp WT1185	3EUKT595038 SEQ ID NO: 8	73	85
8	<i>Picochlorum celeri</i>	3EUKT2133049 SEQ ID NO: 9	44	62
9	<i>Tetraselmis</i>	3EUKT668672 SEQ ID NO: 10	73	82
10	<i>Ostreococcus lucimarinus</i>	XP_001417471.1 SEQ ID NO: 11	71	86
11	<i>Micromonas commoda</i>	XP_002507884.1 SEQ ID NO: 12	72	87
12	<i>Arabidopsis</i>	AT1G12910 SEQ ID NO: 13	70	83

An alignment study of the protein sequences from the Table 2 orthologs was conducted and showed a high degree of sequence conservation across Chlorophyte algae species and across the entire length of the proteins, and particularly in key domains as illustrated in FIG. 5. Sequence similarity was calculated using the bioinformatics matrix BLOSUM62 for sequence alignment.

It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention.

The invention illustratively described herein suitably may be practiced in the absence of any element or elements,

limitation or limitations which is not specifically disclosed herein. Thus, for example, in each instance herein any of the terms “comprising”, “consisting essentially of” and “consisting of” may be replaced with either of the other two terms. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention that in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the

scope of the invention claimed. In addition, where features or aspects of the invention are described in terms of Markush groups, those skilled in the art will recognize that the invention is also thereby described in terms of any individual member or subgroup of members of the Markush group. For example, if X is described as selected from the group consisting of bromine, chlorine, and iodine, claims for X being bromine and claims for X being bromine and chlorine are fully described. Other embodiments are within the following claims.

SEQUENCE LISTING

Sequence total quantity: 26
 SEQ ID NO: 1 moltype = AA length = 340
 FEATURE Location/Qualifiers
 REGION 1..340
 note = misc_feature - WD40 repeat protein, 3EUKT2027500
 source 1..340
 mol_type = protein
 organism = *Oocystis* sp.
 SEQUENCE: 1
 MAEPDANGAS DGRKAEIYTY EFPNLYVSMN WTSRRDKKFR LAVGSFIEDY NNVVNIISLD 60
 EEQKGFVCDP SLTFKHPYPP TKVMFVPDRE GTRPDLLATT GDYLRVWKIG EDGVTLQKLL 120
 NDNKNSEFCA PLTSPDWNET DPKRLGTSSI DTTCTIWDIE KGVVDTQLIA HDKEVYDIAW 180
 GGVGVFASVS ADGSVRVFDL RDKEHSTIY ETPSPETPLL RLGWNKQDPR YMATIVMDSN 240
 RVVVLDIRVP TVPVAEQQRH QACANALAWA PHSSCHICTA GDDAQALIWD LSAVSKEGDS 300
 GLDPILAYNA GQEVNQLQWS STQPDWVAVC FGNKAQILRV 340
 SEQ ID NO: 2 moltype = AA length = 335
 FEATURE Location/Qualifiers
 REGION 1..335
 note = misc_feature - WD40 repeat protein
 source 1..335
 mol_type = protein
 organism = *Coccomyxa* sp.
 note = *Coccomyxa* subellipsoidea
 SEQUENCE: 2
 MDGRLNDRRA EIITYDSENI VYGLSWSNRR DKKFRLAVGS FIEEYDNYVE IITLDDATCK 60
 FTSDAQLAFQ HPYPTKIMF MPDKEGAQPD LLATTGDYLR IWQLKEDGTQ LVKLLNNNKN 120
 SEFCAPLTSF DWNETDLNRL GTSSIDTTCT IWDIEKGVVD TQLIAHDKV YDIAWGGGVV 180
 FASVSADGSV RVPDLRDKEH STIIYDSPQP DTPLRLGWN KQDPRYMATV LMDSTKVVIL 240
 DIRYPTLPVA ELQRHQAPVN AVAWAPHSSC HICSAGDDAQ ALIWLSSMS RPMQDTLDPI 300
 LAYSAGAEVN QLQWSTQPD WVAICFANKT QILRV 335
 SEQ ID NO: 3 moltype = AA length = 359
 FEATURE Location/Qualifiers
 REGION 1..359
 note = misc_feature - WD40 repeat protein
 source 1..359
 mol_type = protein
 organism = *Chlamydomonas reinhardtii*
 SEQUENCE: 3
 MSASDKRERQ EVITYVAPDP VYAMNWSVRR DKRFLGVAS FREDVTNYVD IVSLDDDESDE 60
 LRADPGLRFP HDYPATKLMW MPDREGCRPD LLATTGEALR IWRVCDGSEG EESGSGPGGR 120
 GVQLRSLN LN KQSEFSAPL TSFDWNEADP KRLGTSSIDT TCTIWDIEKG EVDTQLIAHD 180
 REVYDIAWGG LGVFATVSAD GSVRVFDLRD KEHSTIYES PQPDTPLRL GWNRQDPRYM 240
 ATILQDSPKV VILDIRYPTL PVAELCRHQA PVNALAWAPH SAQHICTAGD DSQALIWDVS 300
 AVGGGNANANA AAGGGASDVS LDPILAYGAA SEVNQLQWSS AQPDWVAICF GNKTQILRV 359
 SEQ ID NO: 4 moltype = AA length = 351
 FEATURE Location/Qualifiers
 REGION 1..351
 note = misc_feature - WD40 repeat protein
 source 1..351
 mol_type = protein
 organism = *Volvox* sp.
 note = *Volvox carteri*
 SEQUENCE: 4
 MSNSDKRAEI YTYVAQDPVY AMNWSVRRDR RFRLAVGSFR EDVTNYVEII SLDDDAEELR 60
 SDPSLRFHHD YPATKLMWLP DREGCRPDLL ATTGEALRIW RVLDPDSVAG DGEDLRALLN 120
 NNKQSEFSAP LTSFDWNEAD PKRLGTSSID TTCTIWDIEK GEVDTQLIAH DREVYDIAWG 180
 GLGVFATVSA DGSVRVFDLR DKEHSTIYE SPQPDTPLLR LGWNRQDPRY MATILMDSPK 240
 VVILDIRYPT LPVAELHRRH APVNALAWAP HSAQHICTAG DDSQALIWDV SAVSGGGGQP 300
 GALGGGTAGD VSLDPILAYG AQSEVNQLQW SSAQPDWVAI CFANKTQILRV 351
 SEQ ID NO: 5 moltype = AA length = 341
 FEATURE Location/Qualifiers

-continued

REGION 1..341
 note = misc_feature - WD40 repeat protein
 source 1..341
 mol_type = protein
 organism = *Chlorella protothecoides*
 note = *Auxenochlorella protothecoides*

SEQUENCE: 5
 MSGPSGDKRA EIYTHESADP IYALNWSVRT DKPFRLATGS YVEDQNNHID IIVLDEAREQ 60
 FQADPRLSFV HPFPATKLMF LPVKDPNQPD LLATTSDFLR IWSISEGVA LEKLLNNTKT 120
 SEYCEPITSF DWNHLEPRRL GTASLDATCT VWDIERGCVD TQLIAHDGEV YDLAWGGATM 180
 FASVSADASV RVFDLRDRDH STITYESRGG EALVRLGWNR ADPRFMAVLA AGSAEVVVLD 240
 VRRPAAPLAR LARHTAPANV LAWAPHSACH LCSAGDDGAA LIWDVGALGG GGGPGGAAQD 300
 PGLDPILAYN AGAEVAALQW SAAQPDWVAI AFGNNAQVLR V 341

SEQ ID NO: 6 moltype = AA length = 351
 FEATURE Location/Qualifiers
 REGION 1..351
 note = misc_feature - WD40 repeat protein
 source 1..351
 mol_type = protein
 organism = *Chlorella sorokiniana*

SEQUENCE: 6
 MQQQGEGRAE IYTYESPHLV YGAGWSVRPD KPFRALGSGF IEDYANRVEI VQLDEGRGVI 60
 RSNPALGFQH PYPPTKVGF I PDKDGTTRPDL LATSGDFLRL WRIHDEPGSN QHVRLEKLLN 120
 NNGGGEFSAP LTSFDWNELD PRRIGTASID TTCTVWDVER GVVDTQLIAH DKEVYDIAWG 180
 GVGIFASVSA DGSVRVFDLR DKEHSTIYE SPQPSTPLLR LSWNKQDPRY IAAFAMDSSK 240
 VLVLDIRYPT LPVAQLQRHQ ASVNAVWAP HSAVHLCSAG DDCQALIIDL ALSGAMGGQQ 300
 QDGTAAAAA GGLDPILAYN AGTEINQLQW SASQPDWVAI CFGNKAQILR V 351

SEQ ID NO: 7 moltype = AA length = 355
 FEATURE Location/Qualifiers
 REGION 1..355
 note = misc_feature - WD40 repeat protein
 source 1..355
 mol_type = protein
 organism = *Chlorella* sp.
 note = *Chlorella variabilis*

SEQUENCE: 7
 MQDQQQQGEG RAEIYTYSSS ASVYACGFSS RPDKPFRLLAV GSFIDDYANK VEIIQLDEAA 60
 GVVRRNPALT FQHPYPTKV AFIPDKSGTR PDLLATSGDF LRLWRVSDPE GAQQGVRLK 120
 LLNNNKGDF AAPLTSFDWN ELDPRRVGTA SIDTTCTVWD VERGVVDTQL IAHDKEVYDI 180
 AWGGVGIFAS VSADGSRVVF DLRDKEHSTI IYESPQDPT LLRLSWNKQD PRYIAVLAMD 240
 SPRVTVLDIR YPTLPVLAEL RHQAGVNAIC WAPHSATHLC SAGDSSQALI WDLGLLGTLG 300
 QQPEGGPPGA AAAGGGLDPI LAYNAGAEVN QLQWSPAQPD WVAICFGNKT QLLRV 355

SEQ ID NO: 8 moltype = AA length = 334
 FEATURE Location/Qualifiers
 REGION 1..334
 note = misc_feature - WD40 repeat protein
 source 1..334
 mol_type = protein
 organism = unidentified
 note = *Parachlorella* sp.

SEQUENCE: 8
 MQRAEIHTEY SPTLVYALNW SVRPDKPFRLL AIGSYIEDYN NRVEIVTLGE DGNMGRPSPR 60
 HTFQHPYPPT KLQFVPDPDG SRPDLASSG DFLRLWRITE DGVSLEKLLN NNKASEFCAP 120
 LTSFDWNEND PKRVGTASID TTCTVWDIEK GVVDTQLIAH DKEVYDIAWG GVGVFASVSA 180
 DGSVRVFDLR DKEHSTIYE SPQPDTPLLR LAWNKQDPRY MATTALNSSA IIVLDIRFPT 240
 VPVVELSKHQ AACNAVAVAP QSANHICSAG DDCQALIIDL STLGEAGAGQ AGSPPLDPIL 300
 SYMAGAEVNQ LQWSASHPDW VAICFGNKTQ ILRV 334

SEQ ID NO: 9 moltype = AA length = 337
 FEATURE Location/Qualifiers
 REGION 1..337
 note = misc_feature - WD40 repeat protein
 source 1..337
 mol_type = protein
 organism = unidentified
 note = *Picochlorum celeri*

SEQUENCE: 9
 MEHPPAPNIL TYDSSSIVFA LDWSSRQDKG VRVAVGSFVE GVSNTVEILR VTPAGLIVDD 60
 KETFGIEYPA TQVGFIPTDR CNKPDLLATS GDAVRLWKIS DAGTTLELVL NDPKNTSKNF 120
 SAVTCFDWSE INVKVLAAQS SAGRLLLDWT ESGRLQGTMV GHEDEILDQC WAANDVIVSS 180
 SGDGSIKMYD LRDKDHCTVL YETPRRTFVP RFCWNKLDPR HLAFSIEKSR LVSVDLVRFP 240
 TEPVILLDGH MGNCTALGWS PHREEYLCVS GDDCHALIWD VGKVNSEEDS KPNREAVDAS 300
 PILAYNAQAE INAMAWNPID PDWIAICARN RTQVLRI 337

SEQ ID NO: 10 moltype = AA length = 412
 FEATURE Location/Qualifiers

-continued

REGION 1..412
note = misc_feature - WD40 repeat protein
source 1..412
mol_type = protein
organism = Tetraselmis sp.

SEQUENCE: 10
MPTSDATQAH EHHHTLAATP TQQANNAAPL ADRFTLGLLA MASGPEDRGA GAAGAAPHQR 60
GDSNGKAVTD KRGEIYTYEA PYPVYGMNWS VLLQVREDMK FRLAVGSFVE DVENAVELIR 120
LNEETGKFES NPAHKFVHPY PPTKIMFIPD RDCSRPDLLA TTGDYLRLLWR VEEDGVTLHK 180
LLTNKNSEF CAPLTSFDWN EADPRQLGTS SIDTTCTIWD IERGVVDLQL IAHDKEVYDI 240
AWGGQGVFAS VSADGSRVRF DLRDKDHSTI IYESGMPEIP LLRLGWNKQD PRYMATILMD 300
SSKVVVLDIR YPTMPVAELE AHHKPVNALA WAPQSSSHIC TAGDDAQUALI WNLAPMGQTG 360
PMGGAAPAVL GADLDPILAY NAGEEINQLQ WSTQSDWVG ISFGNKIQIL RI 412

SEQ ID NO: 11 moltype = AA length = 332
FEATURE Location/Qualifiers
REGION 1..332
note = misc_feature - WD40 repeat protein
source 1..332
mol_type = protein
organism = unidentified
note = Ostreococcus lucimarinus

SEQUENCE: 11
MNAEKRAEII TYEAPWMIYA CNWSVRQDKR FRLALGSFVE EYSNKVEIIT LDEETGEFPK 60
EAQCSFTHPY PCTKILFIPD KECTKEDLLA TTGDYLRIRW VQDDNTVQMK SLLNNKSSE 120
FCAPLTSFDW NETKLQVRGT SSIDTTCTIW DIERECVDTQ LIAHDKEVYD IAWGGPEVFA 180
SVSADGSRVF FDLRDKDHST IYESQTPDT PLLRLGWNKQ DPRYMATICM DSPVIILDIR 240
FPTLPVAELQ SHRASVNTLA WAPHSSSHMC TAGDDSQALI WDLSSMNQPP EGGLDPILAY 300
SAGAEINQLQ WSASQPDWIS IAFRNSLQIL RV 332

SEQ ID NO: 12 moltype = AA length = 336
FEATURE Location/Qualifiers
REGION 1..336
note = misc_feature - WD40 repeat protein
source 1..336
mol_type = protein
organism = Micromonas sp.
note = Micromonas commoda

SEQUENCE: 12
MAAMGSGQSG AEIYTYEAPW LVYAMNWSVR QDKRFRLALG SFVEEYSNKV EIITLDEQRR 60
EPPAEPTHRF DHPYPCTKIM FVPDAEGTSE DLLATSGDYL RVWRIGDDGV HLRSLNNNK 120
NSDFCAPLTS FDWSTTNLAR VGTSSSLDTTC TIWDLKETV DSQLIAHDKE VYDIAWGGPE 180
VFASVSADGS VRVFDLRDKE HSTIVYESPT PDTPLRLGW NKQNPYMAT MEMDSAKVVV 240
LDIRVPALPV AELKKHRAAV NTLAWAPHSS RNICTAGDDA QALIWDLSSV AQPGEDEGMDP 300
MLAYNAGAEI SQLQWSATQT DWTAIAFGKN LQVLHV 336

SEQ ID NO: 13 moltype = AA length = 346
FEATURE Location/Qualifiers
REGION 1..346
note = misc_feature - WD40 repeat protein
source 1..346
mol_type = protein
organism = Arabidopsis sp.

SEQUENCE: 13
MGTSSDPIQD GSDEQQRKSE IYTYEAPWHI YAMNWSVRD KKYRLAITS L LEQYPNRVEI 60
VQLDESNGEI RSDPNLSFEH PYPPTKTIFI PDKECQRPDL LATSSDFLRL WRIADDSRV 120
ELKSCLSNKN NSEFCGPLTS FDWNEAEPRR IGTSSSTDTC TIWDIEREAV DTQLIAHDKE 180
VFDIAWGGVG VFASVSADGS VRVFDLRDKE HSTIYESSE PDTPLVRLGW NKQDPYMAT 240
IIMDSAKVVV LDIRFPALPV VELQRHQASV NAIWAPHSS CHICTAGDDS QALIWDISSM 300
GQHVEGGLDP ILAYTAGAEI EQLQWSSSQP DWVAIAFSTK LQILRV 346

SEQ ID NO: 14 moltype = DNA length = 12
FEATURE Location/Qualifiers
misc_feature 1..12
note = partial sequence of WD40 repeat protein
source 1..12
mol_type = unassigned DNA
organism = Oocystis sp.

SEQUENCE: 14
agactcgac cg 12

SEQ ID NO: 15 moltype = DNA length = 1643
FEATURE Location/Qualifiers
misc_feature 1..1643
note = WD40 repeat protein, 3EUKG2027500, encodes SEQ ID NO: 1
source 1..1643
mol_type = unassigned DNA
organism = Oocystis sp.

-continued

SEQUENCE: 15

atggccgagc	cggaacgcga	cggcgcgtcg	gatggcaagc	gcgcggagat	ttacacgtac	60
gagttcccca	acctcggtta	ctccatgaac	tggacggtaa	gcaagcacag	ttgtagcgca	120
cacagccttg	ggtgctgctg	cgtccttgct	gatggtcctg	cttgcgctcg	ccacccgttt	180
tctcacctac	cgaaactggc	tgcgcacccc	tttcagttat	gcttgtaatc	accgcctcat	240
tacctttgtg	tgcagtctcg	tccggacaag	aagtttcgac	tggcagtggtg	cagcttcctc	300
gaggactata	ataacgtcgt	caatatcata	tccgttacgc	gccttccccg	cactaccacg	360
cgtagcggct	cagtctgtcg	tgcggggttg	gcacgacagt	cgttgggttg	aagactattt	420
gaactattgc	tgatttagta	cgcccatgct	ctctcgtaaa	cgccggcgga	tggtcccttt	480
cgcccccact	accatctggc	cgcgacctgt	ggctctgacg	cctcgctggg	tctcgacctg	540
ctgaccatac	tccctcgtaa	ctggctgcaa	cgcagtggac	gaggaacagg	gcaagtttgt	600
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gaagatcggt	gagatggggc	tcacgctgca	gaagctgctg	aacgatgtaa	ggcctagatt	780
aacgtccagc	gctgtgggga	aggaccgacg	gcacggggcg	gaaagggaca	tgcatgccgt	840
accgggggtg	atcacggggc	acggcccacc	ggatcatcat	cttcctcgct	tccaccaacc	900
ctgcgcaacg	tccctcgcaa	cgtcttgaac	aatattttgc	atcttttcac	ttcatccatc	960
ctcgtcatgg	cgaaattaac	ttgcagaaca	agaacagcga	gttttgccgc	ccgctcacat	1020
cgttcgactg	gcacgagacc	gacccaagc	gcctgggcac	cagctctatc	gataccacgt	1080
gcacgatctg	ggacatcgag	aagggcggtg	tggacacgca	gctcatcgcg	cacgacaagg	1140
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cggtgcgagt	cttcgacttg	cgagacaagg	agcatagcac	gatcatctac	gagacgccgt	1260
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tatccaagga	ggcgagctcg	ggcctggacc	ccatcctcgc	gtacaatgca	ggtcaagagg	1560
taaacccagc	gcagtgggtc	tcgacgcagc	ccgattgggt	ggcagctctg	tttggcaaca	1620
agggcgagat	cctgcgagtg	tga				1643

SEQ ID NO: 16 moltype = DNA length = 1023
FEATURE Location/Qualifiers
misc_feature 1..1023
 note = WD40 repeat protein, cds of 3EUKG2027500, encodes
 SEQ ID NO: 1
source 1..1023
 mol_type = unassigned DNA
 organism = Oocystis sp.

SEQUENCE: 16

atggccgagc	cggaacgcga	cggcgcgtcg	gatggcaagc	gcgcggagat	ttacacgtac	60
gagttcccca	acctcggtta	ctccatgaac	tggacgtctc	gtcgggacaa	gaagtgttca	120
ctggcagtg	gcagcttcat	cgaggactat	aataacgtcg	tcaatatcat	atccttggac	180
gaggaacagg	gcaagtttgt	atcgacaccg	tcactgacct	tcaagcatcc	gtaccaccg	240
accaaggtga	tgtttgtgcc	agaccgggaa	ggcactcgcc	ccgacctgtt	ggccaccacc	300
ggcgactatc	tgccgctgtg	gaagatcggt	gaggtatggc	tcacgctgca	gaagctgctg	360
aacgataaca	agaacagcga	gttttgcgcg	ccgctcacat	cgttcgactg	gaacgagacc	420
gaccccaagc	gcctggggc	cagctctatc	gataccacgt	gcacgatctg	ggacatcgag	480
aagggcggtg	tggacacgca	gctcatcgcg	cacgacaagg	aggtgtatga	catcgcggtg	540
ggcgccgctg	gcgtcttcgc	gcccagcggt	cggtgcgagt	cttcgacttg		600
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agggcggtgc	caaacgcgct	tgccctgggg	ccgcacagca	gctgccacat	ctgcacagcg	840
ggcgacgacg	cacaggcgct	gatctggggc	ctcagcgccg	tatccaagga	ggcgagctcg	900
ggcctggacc	ccatcctcgc	gtacaatgca	ggcacaagag	taaacagct	gcagtgggtc	960
tcgacgcagc	ccgattgggt	ggcagctctg	tttggcaaca	agggcgagat	cctgcgagtg	1020
tga						1023

SEQ ID NO: 17 moltype = DNA length = 4908
FEATURE Location/Qualifiers
misc_feature 1..4908
 note = WD40 repeat protein
source 1..4908
 mol_type = unassigned DNA
 organism = Chlamydomonas reinhardtii

SEQUENCE: 17

atgagcgcca	gcgacaagcg	cgagcggcag	gaggtttata	catatgtggc	accggacccc	60
gtgtacgcca	tgaactggag	cgtgagttgc	cctcgggcac	ggctatgacg	ggcgctcgct	120
gggctgctgg	ggacggggct	tgggaaggcg	ggaggcccgc	cgccgtctgc	agaccctgcc	180
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gaatgccatg	agccaagggg	cgctgtggag	aggcagagca	gagggcgggg	atgccaaagca	300
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ccgcaccgcg	aaccatacct	accacaggta	cgccgggaca	agcgcttcgc	actggggcgt	600
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acacggggcg	attgagtaac	atggggattg	cggacggcag	ttacactggt	acgggggtat	720
gaagtggagc	tgtagagtac	agccggagat	aagcacatcg	ttcggcacag	agggagtgtg	780

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cggcatcccc	catcacaaat	gcaggtgaac	cagctgcagt	ggagctcgcg	gcagccggac	4860
tgggtggcaa	tctgcttcgg	caacaaaacg	cagatcctaa	gggtgtga		4908

SEQ ID NO: 18
 FEATURE
 misc_feature
 source
 mol_type = DNA length = 5770
 Location/Qualifiers
 1..5770
 note = WD40 repeat protein
 1..5770
 mol_type = unassigned DNA
 organism = Coccomyxa sp.
 note = Coccomyxa subellipsoidea

SEQUENCE: 18

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ctgaatgaca	gacgggcgga	gatatacacc	tacgactctg	agaacatcgt	atatggcttg	240
agctggagtg	taaggatcca	ttcagtcctat	cttcctggaat	ttaatgtctg	cattatgtat	300
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gactgggtgg caatctgctt cggcaacaag acccagctgc tgcgggtgtg aggcgcgtgc 2820
cgacagagca ccacacgcgc gcgctgctcg ccggctgcag cgttgcctgc cgctccctc 2880
cagggcagcg gcccgccgc gcgcttccg ctcccaagc tgcagcctt cttgctgccc 2940
ttattcgctt gctccccctg tcttgcttcc gctcctgctg taactgcccc cgcccgcctc 3000
tgtatcaccg ggggtgcttt tcttactgc acacgtacca cgcctctgtt ggcaccctgc 3060
ccctccctaa tgcacgcagg tctggcagcg tgcagcccc tctaatgcct gggccctgcc 3120
attcacacca atcgcatcaa ccgtacatct gtccccctgc actgctcttt gtcaccattc 3180
cacctgatgc tctctcctct cccaacccaa ctgatcccc cgtcgcctcc ataccgattt 3240
cgagagacac cttgcaatga aacacgcagc gc 3272

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SEQ ID NO: 21      moltype = DNA length = 3032
FEATURE            Location/Qualifiers
misc_feature       1..3032
                    note = WD40 repeat protein
source             1..3032
                    mol_type = unassigned DNA
                    organism = unidentified
                    note = Parachlorella sp.

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SEQUENCE: 21
atgcagcgcg cagaaatcca tacgtacgag agccccacgt tggtttatgc actcaactgg 60
agttaagtc acacaatatg ttgacaagat actgaagcgc gacatgtata ggctgtgtcc 120
tgaagaggca agctttcacc tgctgtaggt gcggcctgac aaaccttca gactagccat 180
cggaagctac atcgaggact acacaacccg agtggaaatc gtcacacgta tgttcaccgc 240
ttcagggtctt gctgctattg cgttcgtgccc ttcaaatgca gactttcgaa gcaagtttgc 300
ccaccggggg tctctgtgtg agtcggtgaa gatggaaatg gaatgcggcc tagccccagc 360
cacacctttc agcatcccta tccaccaccc aaacttcagt ttgtaccaga tctgatggc 420
tccccggcctg atctgttagc cagctccggt gacttccctc gactctggcg catcacggag 480
gacgggggtt ccctggaaaa gcttctcaac aacgtgagcg cgctctgctc gatagcgctg 540
tctgttgtga ccatggacgg ctagcgacac gcggtagcgc gcagtgcac gaagacgacc 600
ggggctgacg ctactctgaa tgcacacccc ttgctgctg tgggtgcagaa caaagcaagc 660
gagttttgcg cgcccttgac cagctttgac tggaaatgaga acgaccccaa gagggtgggc 720
actgccagca tcgacacacc gtgcactgta tgggacatag agaagggggt ggtggacacc 780
caggtgggtt gagtggagta gagtggagtg gcggagaacc tggaaagaca ggccatgacc 840
agggcatcg tgcagacac tctgcgcgc gccgcgctc cgtcaagtct tggccgttca 900
aagactcgag gtccttccgg tgtgctgctt ggagtcgccc gcctcctgcc acatcgtgtg 960
cctgctgccc ctttggagcg atgctccctc tgccacgcgc tcaccttgg aaccgtgtgg 1020
cccttgacg tgattgcccc tgacaaggag gtgtacgaca tagcatgggg gggcgtgggg 1080
gtgtttgctc cgttgcaagc agatggctcg gtgcgggtgt ttgacttgag gtagggtgctc 1140
tgctgcggcg cttcgacctg gggctgagtg accgggtggg gccggcctgg aggatggagg 1200
tgggaatatg tgcagtagc cgaagtagat gatcttggcg gtggctcatc ctgggtgtggg 1260
tcgctggtag aagaagcgcg gtggaacagg ttaggtgtgt gggcttggag cagttaggtg 1320
caagccagtt gtagagtag cccacgacgc ctcccaaaag ggccctcagc ccgcagcacc 1380
ctgcagcagc gagcttcccc agttgctgcg ccaggggcag cgcctgctcg ggcgcacagc 1440
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cccgacccca cccctcccc ccttctgccc gctgcctcat gcgccctcca ccaacattcc 1680
ctgcagggat aaggagcact cgaccatcat ctatgagtcg ccgcagcccc acacccccct 1740
gctgcgctg gcctggaaca agcaggaccc gcggtacatg gccaccactg ccttcaactc 1800
ctccgcatt gtgggtgctg acatccgatt cccacgggtg ccggtggtgg agctgtccaa 1860
gcaccagggt cgaatgggca gcgggggtgc agtgcctgcg ggggggtgct gcaatggggc 1920
agctgctatc tgccgccttg ctgtgagctg gcctggcgct gggagccagg agtgcctacc 1980
ctgctgcagc cgtgccaagc ggttgctgctg atcgacagc ctgagggccc cgggctgggc 2040
tggaaaggcg ggtgctcacc cggctcaggg gccacgagg atcgccgatt tctgcccctg 2100
gcctggtgat catgcgctg gcggcggtgc ggtggcgtgc gcagcatgcg gctgcgcact 2160
gcattggcca gccacgctg tctgtccatg gtgtgcctgc agcccgacc agccaaagcg 2220
tgctgcgct ctgcctcatg gaacagacct ctcaacgcgc tccctccctc tgtggctagt 2280
gtgccccagc accaggtggc agctggagcg cgccagcat ggtgcagcaa caggaggaaa 2340
gcgcgcgccc ctgggcctgc gggcaatttt agcgcgttgc gctgcgctgc attcctcgca 2400
acgtgcacgt caccgaagcg tgtatctccc ccttcccca ctgctgcag gccgctgca 2460
atgctgtggc ctgggcccc caaagtgcga accacatctg cagtgcgggg gacgattgcc 2520
aggtggggcg cgtggaagg ggggaaggaa agggggaggg gggctcgtgt aactggggcg 2580
aatgctggcc ttctaataa accgccacca acaagctggg gctgtgctt gggaggggcg 2640
gggcagtcgt ccccgcgccc acggggggcg acaccttgc cttcccttgg cgagtattcg 2700
agcacgggac ccttctctcc cctcctgtgc atctcatgtg atgtactcgc cgagccctct 2760
gccccctgct cgactgtcgc gcagctgcaa gggcacgcgc caagcgcatt gcggcgctgt 2820
ccctccgcgc cctggtgcgg tgcgtgcagg ctttgatctg ggacctgtcc actctggggg 2880
agggcgcgcg gggccaggcg gggagcccc ccttgagccc catcctgtcc tacatggcgg 2940

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gggaggaggt gaaccagctg cagtgggtcg cgtcccaccc cgaactgggtg gccatctgct 3000
ttgggaacaa gacgcagatt ttgagggtgt ga                                     3032

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SEQ ID NO: 22      moltype = DNA length = 1045
FEATURE           Location/Qualifiers
misc_feature       1..1045
                   note = WD40 repeat protein
source            1..1045
                   mol_type = unassigned DNA
                   organism = unidentified
                   note = Picochlorum sp.

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SEQUENCE: 22
atgtcgactg attggttggg tgtaatgtg cagtctaggc aagataaggg tgtacgagta 60
gctgttggca gttttgtgga ggggtgtctg aacacagtcg agattcttcg cggtaacggat 120
catctgtgag tctctgttct acttgggaaa tgtgggcata attttttgta tatgagggtgc 180
agtgaactcc gcagggttaa ttgtggatga caagaaaacg ttcggcatag agtatccggc 240
gacgcagggt ggattttatc ctgataggtt ttgcaacaag ccagatttgt tggcaacctc 300
tggggatgct gttagggtgt ggaaaaattc agacgcaggg acgacgcttg aactggtgtt 360
gaatgatcca aagaatacct ctaaaaattt cagtgcgggtg acctgctttg attggagtga 420
aatcaatgtg aaagtgttgg cggcagggtc gagtgcaggg cgattattgc tgtgggacac 480
cgagtcaggg aggtctcagg gcacaatggt gggacatgag gatgagattc ttgattgtca 540
gtgggcagct aatgatgtga ttgtttcttc ttcgggcgat ggatcaattc ggatgtatga 600
cctgcgggat aaagaccatt gcacggtatt gtatgagacc ccaggaggga cccctgtgcc 660
gaggttttgt tggaaacagc tggatccgag gcattcttga tttctctatg aaaagagtcg 720
gcttgtagt gtctcagat ttctgcttcc gacagagccg gtgatcttgc tggacgggtc 780
tatggggaac tttacagcac ttgggtgggc cctcacaga gaggaatacc tctgctcggt 840
tggagatgat tgccatgcat tgatatggga cgtggggaag gtgaatagtg aggaggatag 900
taagccaatc cgagaggcgg ttgacgcac tcctatccta gcgtacaatg ctcaggcaga 960
gatcaatgct atggcttgga atccaataga cccagattgg attgccattt gcgctagaaa 1020
cagaacacaa gtatttgaaa tatga                                     1045

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SEQ ID NO: 23      moltype = DNA length = 2857
FEATURE           Location/Qualifiers
misc_feature       1..2857
                   note = WD40 repeat protein
source            1..2857
                   mol_type = unassigned DNA
                   organism = Tetraselmis sp.

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SEQUENCE: 23
atgccgacga gcgacgcgac gcaggcacat gaggcaccat acgtgccgca cgcgcggccg 60
cgacccccgc caagaacggt gtctgcatgt gccatgttga tgagcaagga gaaacggcgg 120
cgccacgcag acaagagctg gccgattttt tggccagatg gtgtgtgggt agctttcgcg 180
agctccacca agaagcctc agcgtcgggg ccagctcgtc ttgcataatg cttcggcaat 240
ttgattgttg actagcgcct acagtacatg tacaaacgag cgtcgctac ggggtatacgg 300
agggcttgcc gtaactgtgg atcgctgcct attaaacggc cgtcgtaaaag atagacgctg 360
caagggtttc actggtgtca gtctgttccc gggggcggga aaacaaaaac cctggtcgta 420
aatgagagct ggcaaaagttt caccgcgcgc agtcgcgtat ccaggaaaaa aaaacccgcc 480
cctcgacact tctcgtggcc acaggtagag gactgccgaa gtgatcacc caccagaaag 540
cccaccgtcg ccagcgctac cgtttacgtg cgggtactca gactctcgcc gcgacacca 600
cgcaacaagc caacaacgcc gccccgcttg cggatcgctt caccgcttgg ctggtgcgga 660
acctgtttca gggaccttgt tcccgccgac aaattattgc tgcctcgctc tccctgactc 720
cgacagcgct gcaccaggct ggcctgttca ccgcccgggg ccggtgtggg ttgtggccca 780
cgcgagtgcc cactcgggtt cctttcagcc tacgctgggc gtaaaagcct taccagcatg 840
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ttgtggcgcc aggtcgggtg ctgtgccccg ggcgacgcgc gcctgggttg gctagcttgc 960
gatggcgagc gggccggagg accgggggtg cggggcgggc gggggcgggc cccaccaacg 1020
agggcatagc aacggcaaaag cgggtgacaga caagcgcggg gagatataca cctacgaggg 1080
gccgtaccgc gtatacggga tgaactggag tgtgcgtgcg ccgacatgag ccaagggggg 1140
ccaggggagg gttccccggg gggggggggg gggggaggag tacttggtac aagcagactt 1200
tggccccgtg gctgaggggg tcgagtgttg caggtgcggg aggatatgaa gttccgcctc 1260
gcggtgggaa gctttgtaga ggacgtggag aatgcgggtg agctcatccg gcgtgagtgc 1320
ccgagcgctg acgcaggcct gcttctgtgt tgcaggggag gcgcgggcat ccgcctgatg 1380
acgtgctgtg tgcacagtga accaggaaac cggcaagttt gagagcaacc cggcgacaaa 1440
gtttgtgcac ccgtatccac ccacaaaaat aatgtttatc cccgacggcg actgctcgcg 1500
ccctgacctg ctgcgccacca cgggcgatta tctgcggctg tggcgcgctg aggaggacgg 1560
cgtcacgctg cacaagctgc tgacaaatgt gagcgcggaac aaacctttgt gccgcgcctc 1620
ccccaccac caccagtcct ccttcctctt aaggccctt ctcaagagat accacggcca 1680
gctccagcat gaccccgccc cccctatttc gtcaacatgc acctcccccc tcgagaacaa 1740
aaacagtgag ttttgccgcg cattaacatc ctttgactgg aacgaagccg acccgaggca 1800
actgggcacc tcattccatc acacaacatg cacaatatgg gacatcgagg tggggccaggc 1860
agtcacaagg ccccccctcc cccccctccc gcaaatgccc tcttgcgctg gacatgtcaa 1920
aacgcctgcc tctgtgtgag tccagagagg cgttgttgac acgcagctta tgcctcacga 1980
caaggaggtg tacgacattg cgtggggcgg gcaggcgctc tttgccagcg tgcctcgaga 2040
cggctctgtg cagtggtttg acctccggtg gcctcgcggt tacctgcggc cgcgcgcgac 2100
ccacacctgg cgccgtggct ggcttgcggg gggggggggg gggggggctga cgcgcggcgg 2160
ggctgttcgg caccgcaggg acaaggacca ctcaaccatc atctacgaga gcggggatgcc 2220
cgagatcccg ctgctgcggc tgggctggaa caagcaggac ccgcgctaca tggccaccat 2280
cctcatggac tctcccaagg ttgctgctct ggacatcagg tccgcgcgcc ttgcctgcca 2340
tcacgcaaca tatactgggg gtgtgtcttg cggcgctgac ctgtgatgct gcgcgcgcgt 2400

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gcttgcgtgca ggtacccccac gatgctgtga gctgagctgg agggcgcatca caagcctgtg 2460
aatgcccctgg cgtggggcccc gcagtcctcc tgcacatct gcactgcggg ggacgacgca 2520
caggtgtggcg gggatgcatg ctctgatgca tcacaggaga cagagcgaca ggggggggggt 2580
gagcagggggg ggggggggggg ggaagggggg atgcccgggt gaggcagatg gtggctgact 2640
gttgcatgct gccgccagg cgctcatctg gaaccttgcg cccatgggca cccagggggc 2700
catgggggggt gctgcgcctg cagttctagg cgcggacctg gatcccatcc tagcgtacaa 2760
cgccggcgag gaaatcaatc agctgcagt gtccagcacg cagtcgcact ggggtgggcat 2820
atccctttggc aacaagatcc agattttgag aatctag 2857

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SEQ ID NO: 24      moltype = DNA length = 1431
FEATURE           Location/Qualifiers
misc_feature       1..1431
                   note = WD40 repeat protein
source            1..1431
                   mol_type = unassigned DNA
                   organism = unidentified
                   note = Ostreococcus sp.

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SEQUENCE: 24
cggtcgacgc gctcgacgcg gtcgacgcga gcgaggcgcg cgcttcgagc gacgcgaagg 60
cgtcgggacg gaacgcgaag gcgtcgagcg cgaacgcggg acccgacggg cggtgaaggc 120
gcgcgacgaa agacgaggaa ggcgcgcgcg aacgatgaac gcggagaaga gggcggaaat 180
atacacttat gagcgccgtt ggatgatcta cgctgcaat tggagcgtgc gtggcgagcg 240
aggcgatgga ttggggcgga gcgcgggaga attgaatcg gcggggcgac ggaggagacg 300
cgacggaggga gactcgggga cgcgcgcgaa cggtcgatcg gagattaaaa atggagacgc 360
gcgagtgaag acgcgaatgg cgtggactga cgacgtcgaa ttgaacgcga caggttcgac 420
aagataaacg cttccgcctc gccttgggtt cgttcgtgga ggagtatagc aacaagggtg 480
agatcatcac cttggacgag gaaaccgggg agtttccgaa ggaggcgagcgt ttctcgttca 540
cgcatccgta tccttgacag aaaattttgt tcattccgga caaggagtgc acgaaggagg 600
atctgttagc gacgacgggg gactacttgc gaactctgca agtgcaggat gataaacacg 660
tgcagatgaa atctttactg aataataaca agagcagcga attttgcga ccgctgacga 720
gctttgattg gaacgagacc agccttcagc gagtggggac gtcgtcgatc gacacgacgt 780
gtacgatttg ggacatcgag cgcgagtcg tggacacgca gctcatcgcg catgataagg 840
agggtgtacg catcgcgtag ggtggtccag aggttttcgc tagcgtaagt gcggatggaa 900
gtgtgcgagt ttctgacttg agagacaagg atcacagtac gatcatttac gagagtcaaa 960
ctccagacac gccctgtgtg cgtttgggtt ggaacaagca ggatecgaga tacatggcca 1020
ccatttgcac ggatagtcgc gtgatcattc tcgatattcg cttcccgacg ttgccggtcg 1080
cagaacttca ctggccacga gcgagcgtga atacattggc gtgggcgcca cacagctcaa 1140
gccacatgtg cagggcgggc gacgacagtc aggcgttgat ttgggatttg tcgtccatga 1200
atcaaccacc cgaaggcggt ctgcagcccta ttctcgctta ctctgctgga gcagaatatca 1260
atcagttaca gtggagcgcg tcgcaaccgg attggatctc gatagctttc cgaacacagc 1320
tccagatcct ccgagtttag tcaacgcgct gtcagggtctg cgccgacgac actgtatatt 1380
acccgaattt ccgataacgc gacacacgac acacgacacg cagcgacgta g 1431

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SEQ ID NO: 25      moltype = DNA length = 1294
FEATURE           Location/Qualifiers
misc_feature       1..1294
                   note = WD40 repeat protein
source            1..1294
                   mol_type = unassigned DNA
                   organism = Micromonas sp.
                   note = Micromonas commoda

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SEQUENCE: 25
atggcgggcca tgggcagcgg ccagagcggc gccgagattt acacgtacga ggccgcatgg 60
ctcgtgtacg cgatgaattg gagtgtgagt gcccgtcgat gactotgctc cgtcccgcgc 120
cgttcctccc cgcccggggc gatccctcgc ctgcacccaa tctgaccogg caagatccgc 180
tgctgacgcg accttgagga cgcgcgccgc agtcgacgca cgcgccccgc ccgcgacgt 240
gaccgcgtga cgcttcactc gatataaacc tccccctccc cgcgcgcgca ttcaacaggt 300
gaggcaggac aagaggtttc gcctcgcgct cgggtcgttc gtggaggagt acagcaacaa 360
ggttgagatc atcacgctgg acgagcagcg acgggagttt cggcgaggag cgacgcacag 420
gttcgaccac ccgtaccgct gcacgaagat catgttcgct ccagacgcgc agggaaaccg 480
cgaggactta ctggccacga gcggcgacta tctgcgggtt tggcgcatag gcgacgacg 540
cgtgcacctg cggagcctcc tgaacaacaa caagaacagc gacttttgcg cgcgcgtcac 600
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gagcgtcagg gtgttcgacc tgcgggataa ggaccacagc acgatcgtct acgagtcgcc 840
gacgcccggc acgcccgtgc tgaggttggg ttggaacaag cagaaccgga ggtacatggc 900
gacgatggag atggacagcg ccaaggttgt ggtgctggac attcgcgtgc ccgcgctgcc 960
ggtggcgagg ctgaagaagc acagagccgc ggtgaacacg ctggcgtggg cgcgcacag 1020
ctcgaggaa atatgcaccg ccggggacga cgcgcaggcg ctcatattgg acctgtcgtc 1080
ggtggcgag ccgggggagg acgggatgga tccgatgctg cgtacaacg cggggcgga 1140
gatcagtcag ctgcagtgga gcgcgacgca aaccgactgg atagccatcg cattcgcaa 1200
aaacctgcag gtgcttcacg tgtgacgccc gcggggagaa cgtggcgatc gtatgctcag 1260
ttcggttttg aattcaacgt tcatttagca ctca 1294

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SEQ ID NO: 26      moltype = DNA length = 1041
FEATURE           Location/Qualifiers
misc_feature       1..1041
                   note = WD40 repeat protein

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source          1..1041
                mol_type = unassigned DNA
                organism = Arabidopsis sp.

SEQUENCE: 26
atgggaacga gcagcgatcc gattcaagat ggttcogatg agcagcagaa gcgatcagag 60
atctatacat acgaagcgcc atggcacatc tacgcaatga attggagcgt tcgtcgcgat 120
aagaagtatc gtctcgccat cactagcctc ctcgagcaat acccgaaacc tgctcgagatt 180
gtgcagctcg atgaatccaa tggtagatc cgttcogatc ctaacctctc ctttgagcat 240
ccttatccac caacgaagac cattttcata cctgacaagg aatgccaaag acctgatctt 300
ctcgctactt caagtgattt ccttcgttta tggagaatcg ctgatgatca ttcccgtgtt 360
gagctcaaat cttgtctcaa tagcaataag aacagttagt tttgtggtcc tcttacttct 420
tttgattgga atgaagctga gccacgtcga attggaacat ctagtactga tacgacttgt 480
actatctggg acattgagcg tgaagctgtt gatactcagc ttattgctca tgataaggaa 540
gtttttgata ttgcttgggg tgggtgttgt gtttttgcat ctgtttcagc tgatggctcc 600
gttaggggtg ttgatcttcg tgataaggaa cattcgacga ttatctatga gagctccgag 660
cctgatactc ctttagtgcg tcttggttgg aacaaacagg atcctaggta catggctact 720
attatcatgg acagtgcctaa agttgtgggt cttgacattc gttttccggc tcttctgtg 780
gttgagcttc aacgacatca agctagtgtc aatgccattg cttgggctcc tcatagctct 840
tgtcacattt gtactgctgg agatgattct caagctttga tttgggatat ttcattcatg 900
ggacagcatg ttgaaggtgg tcttgacctt attctagctt acactgctgg tgctgagatt 960
gagcagcttc agtgggtctc ttctcagcct gattgggtcg caattgcttt ctctaactaa 1020
ctgcaaatcc tcagggtttg a                                     1041

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What is claimed is:

1. A method of producing a composition containing lipids comprising,

performing a deletion, disruption, or inactivation in an algal organism in a nucleic acid sequence having at least 90% sequence identity to SEQ ID NO: 15 and encoding a WD40 repeat containing protein having at least 90% sequence identity to the polypeptide of SEQ ID NO: 1 to produce a recombinant algal organism;

culturing the product recombinant algal organism in a culture medium, and thereby producing a composition containing lipids,

wherein the recombinant algal organism has higher lipid and/or higher biomass productivity versus a corresponding control algal host cell not having the deletion, disruption, or inactivation of the nucleic acid comprising the nucleotide sequence having at least 90% sequence identity to SEQ ID NO: 15 and encoding a WD40 repeat containing protein having at least 90% sequence identity to the polypeptide of SEQ ID NO: 1.

2. The method of claim 1 further comprising harvesting the composition containing lipids from the recombinant algal organism.

3. The method of claim 1 wherein the recombinant algal organism has at least 30% greater lipid productivity versus a control alga that does not have the deletion, disruption, or inactivation.

4. The method of claim 1 wherein the recombinant alga exhibits at least 40 grams per square meter per day of lipid production after 5 days of cultivation.

5. The method of claim 1 wherein the deletion, disruption, or inactivation is a frame shift mutation.

6. The method of claim 1 wherein the deletion, disruption, or inactivation is a knock out mutation.

7. The method of claim 1 wherein the deletion, disruption, or inactivation is performed by mutagenesis.

8. The method of claim 1 wherein the culturing is performed under nitrogen deficient conditions.

9. The method of claim 1 wherein the recombinant alga has higher biomass productivity per unit time as measured

by production of total organic carbon (TOC) versus a control alga that does not comprise the deletion, disruption, or inactivation.

10. The method of claim 9 wherein the recombinant alga has at least 15% higher biomass productivity.

11. The method of claim 1 wherein the algal organism is a Chlorophyte alga.

12. The method of claim 11 wherein the Chlorophyte alga is of the class Trebouxiophyte.

13. The method of claim 12 wherein the recombinant algal organism is an alga of the genus *Oocystis*.

14. The method of claim 1 further comprising: mutagenizing a population of the algal organisms to perform the deletion, disruption, or inactivation in an algal organism in a nucleic acid sequence having at least 90% sequence identity to SEQ ID NO: 15 and encoding a WD40 repeat containing protein having at least 90% sequence identity to the polypeptide of SEQ ID NO: 1;

screening the mutagenized algal organisms for higher lipid productivity;

sequencing at least a portion of the genome of the mutagenized algal organisms;

identifying the deletion, disruption, or inactivation in the gene encoding a WD40 repeat family protein compared to the population of algal organisms prior to mutagenesis; and

recapitulating the deletion, disruption, or inactivation in a parental strain of the mutagenized algal organisms to produce the product recombinant algal organism.

15. The method of claim 14 further comprising harvesting a lipid composition from the algal organism.

16. The method of claim 14 wherein the algal organism is a Chlorophyte alga.

17. The method of claim 14 wherein the Chlorophyte alga is of the class Trebouxiophyte.

18. The method of claim 17 wherein the recombinant algal organism is an alga of the genus *Oocystis*.

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